

Bidirectional Meeting Point Assignment and Dynamic Routing for Many-to-Many Demand-Responsive Transit with Simulated Passenger Response

An Jin^a, Tianliang Liu^{a,*}

^aSchool of Transportation Science and Engineering, Beihang University, Beijing 100191, China

Abstract

Demand-responsive transit (DRT) offers flexible mobility in low-density urban areas, but door-to-door service is operationally costly and difficult to scale. Meeting-point-based DRT reduces vehicle detours by consolidating pickups and dropoffs at pre-defined locations, yet existing work assigns meeting points only on the pickup side and ignores passenger heterogeneity in mode choice. We address this gap by proposing a many-to-many DRT framework with bidirectional meeting point sets (M_r^P for pickup, M_r^D for dropoff) and simulated passenger response via single-offer binary logit acceptance with three passenger types and an outside option. The model is formulated as a coupled three-layer system — service offer generation, passenger response, and dynamic dispatch — and solved with an ex-post MILP routing diagnostic for small instances and a rolling horizon Adaptive Large Neighborhood Search (ALNS) heuristic for large-scale deployment. Numerical experiments on stylized scenarios inspired by Chinese suburban conditions show that, under the unconstrained comparison (FullModel 18.3% vs.

*Corresponding author

DoorToDoor 61.0% served share), FullModel achieves 15.1 vkm/trip versus 21.3 vkm/trip (29.1% improvement). A separate matched-coverage diagnostic suggests that the efficiency advantage persists after coverage differences are reduced, but this diagnostic is treated as exploratory rather than as the primary headline estimate. Equity analysis across passenger types reveals a Gini coefficient of 0.1216 in acceptance rates: time-sensitive passengers are systematically disadvantaged (14.4% acceptance) relative to price-sensitive (25.4%) and walk-sensitive (20.2%) passengers. Based on these findings, we derive five scenario-specific policy implications for DRT deployment in low-density Chinese urban areas.

Keywords: demand-responsive transit, meeting points, passenger choice, ALNS, rolling horizon, equity

1. Introduction

Demand-responsive transit (DRT) has emerged as a promising solution for providing flexible, on-demand mobility in low-density urban areas where fixed-route bus services are economically unviable. In China, rapid suburbanisation has created large residential zones with sparse transit coverage, making DRT an increasingly important policy instrument for urban planners and transport authorities. However, the dominant door-to-door DRT model — in which vehicles travel directly between each passenger’s origin and destination — is operationally costly: vehicles accumulate large detour distances serving spatially dispersed requests, and fleet utilization remains low. Meeting-point-based DRT addresses this inefficiency by requiring passengers to walk to and from pre-defined consolidation points (e.g., bus stops,

commercial landmarks), enabling vehicles to serve multiple passengers along shared route segments. The trade-off is clear: meeting points reduce vehicle-kilometres but impose a walking burden on passengers, and passengers who find the required walk unacceptable will reject the service offer and use an alternative mode. Designing meeting-point assignments and vehicle routes while representing heterogeneous passenger acceptance is therefore a central operational challenge for next-generation DRT systems.

Existing literature on meeting-point DRT has made important progress but leaves a critical gap. The Dial-a-Ride Problem (DARP) literature provides a rich foundation for vehicle routing with time windows and capacity constraints [1, 2]. Recent extensions have introduced meeting points into the DARP framework. Cortenbach et al. [3] proposed the DARPmp, the first formulation of DARP with meeting points, using a Tabu Search heuristic to solve instances with single-sided meeting point assignment: pickup locations are flexible (passengers walk to a nearby stop), but dropoff locations remain fixed at the passenger’s destination. This single-sided approach captures part of the efficiency gain but leaves the dropoff leg unoptimised. Crucially, Cortenbach et al. [3] do not model passenger choice: all offered meeting points are assumed to be accepted, ignoring the reality that passengers with low walking tolerance will reject inconvenient offers. On the dynamic side, Wu et al. [4] proposed a rolling horizon framework for online DRT re-optimization, demonstrating that periodic re-assignment of requests to vehicles significantly reduces operational cost under dynamic demand. However, Wu et al. [4] do not incorporate meeting points: their model retains door-to-door service. No existing work combines: (a) bidirectional meeting

point sets (M_r^P for pickup, M_r^D for dropoff), (b) explicit binary logit passenger choice with heterogeneous passenger types and an outside option, and (c) rolling horizon re-optimization in a many-to-many DRT setting. This is the gap the present paper fills.

We propose a many-to-many DRT model with bidirectional meeting point sets. For each request r , the system constructs a candidate pickup set $M_r^P \subseteq \mathcal{M}$ (meeting points within walking radius ρ^P of the passenger’s origin) and a candidate dropoff set $M_r^D \subseteq \mathcal{M}$ (within ρ^D of the destination). The system selects pickup meeting point $m_r^P \in M_r^P$ and dropoff meeting point $m_r^D \in M_r^D$ for each offered request, together with vehicle routing and scheduling; realized passenger acceptance is then simulated. Passenger response is modeled via a two-alternative binary logit model with three passenger types — price-sensitive, time-sensitive, and walk-sensitive — each with distinct utility coefficients over walking time, waiting time, in-vehicle travel time, and fare. An outside option allows passengers to reject all offered bundles and use an alternative mode. The full model is formulated as a coupled three-layer system: (1) service offer generation, which enumerates feasible bundles $\mathcal{B}_r = M_r^P \times M_r^D$ and selects a routing-feasible offer; (2) passenger response, which determines the accepted bundle based on the binary logit acceptance rule; and (3) dynamic dispatch, which routes vehicles using a rolling horizon ALNS heuristic with re-optimization at fixed time intervals. For small realized accepted sets, we use a Mixed-Integer Linear Program (MILP) implemented in Gurobi as an ex-post routing diagnostic; large-scale simulations use the rolling horizon ALNS heuristic. Numerical experiments on stylized scenarios inspired by Chinese suburban conditions quantify the

efficiency and equity implications of bidirectional meeting point assignment.

The contributions of this paper are as follows:

1. A three-layer many-to-many DRT simulation framework with bidirectional meeting point sets (M_r^P, M_r^D) , rolling horizon routing, and binary logit passenger acceptance.
2. A rolling horizon ALNS implementation and ex-post MILP routing diagnostic for studying bidirectional DARP instances with simulated passenger response.
3. Empirical demonstration of a 29.1% per-trip vehicle-km gain under the unconstrained comparison (FullModel 15.1 vs. DoorToDoor 21.3 vkm/trip), with a separate exploratory matched-coverage diagnostic indicating a 35.0% gain after coverage differences are reduced.
4. Equity analysis across three passenger types (Gini coefficient = 0.1216) and five scenario-specific policy insights for Chinese city DRT operators.

The remainder of this paper is organized as follows. Section 2 reviews related work on DARP, meeting-point DRT, and passenger response modelling. Section 3 presents the problem formulation, including the three-layer coupled model and the binary logit choice structure. Section 6 describes the solution methodology: the ex-post MILP routing diagnostic and the rolling horizon ALNS heuristic. Section 7 reports numerical experiments, including ablation studies and sensitivity analysis across demand density, walking tolerance, and fleet size. Section 8 discusses policy implications for Chinese city DRT operators. Section 9 concludes and outlines directions for future research.

2. Literature Review

2.1. Dial-a-Ride and Demand-Responsive Transit

The Dial-a-Ride Problem (DARP) is a variant of the vehicle routing problem with time windows in which a fleet of vehicles must transport passengers between individual origin–destination pairs while respecting time window constraints, maximum ride time limits, and vehicle capacity [1]. Since its formal introduction, the DARP has attracted sustained attention as a model for paratransit, elderly and disabled transport, and, more recently, on-demand mobility services in low-density urban areas [5, 6].

Demand-responsive transit (DRT) extends the DARP framework to a broader operational context in which service is triggered by real-time passenger requests rather than fixed schedules. The growth of smartphone-based booking platforms and GPS-enabled fleets has made DRT increasingly viable as a first- and last-mile solution in suburban and peri-urban areas where fixed-route bus services are economically unsustainable [7]. Comprehensive surveys by Ho et al. [2] and Pillac et al. [8] document the rapid expansion of DRT research, covering exact methods, metaheuristics, and online algorithms for both static and dynamic problem variants.

A defining characteristic of classical DARP formulations is the assumption of door-to-door service: vehicles travel directly to each passenger’s origin and destination address. While this maximizes passenger convenience, it generates highly fragmented routes with significant detour and deadhead mileage, particularly when demand is spatially dispersed [9]. The present paper relaxes the door-to-door assumption by introducing bidirectional meeting point sets, allowing both pickup and dropoff locations to be selected from a

pre-defined network of accessible stops. This structural change is the central departure from classical DARP and motivates the literature reviewed in the following subsection.

2.2. Meeting Points and Virtual Stops

The concept of meeting points — fixed or semi-fixed locations where passengers walk a short distance to board or alight — was introduced into the ridesharing literature by Stiglic et al. [10], who demonstrated that allowing passengers to walk to nearby virtual stops substantially reduces vehicle detour and increases the number of feasible matches in a shared-ride system. A follow-up study [11] quantified the trade-off between walking burden and system efficiency, showing that even modest walking tolerances (200–500 metres) can yield significant reductions in total vehicle kilometres travelled without materially degrading passenger experience.

Building on this foundation, Cortenbach et al. [3] introduced the Dial-a-Ride Problem with Meeting Points (DARPmp), the first formal DARP extension to incorporate meeting point selection as a decision variable. Their formulation allows the pickup location to be chosen from a candidate set while the dropoff location remains fixed at the passenger’s destination address — a *single-sided* meeting point design. The problem is solved with a Tabu Search metaheuristic applied to a static batch of requests. Cortenbach et al. [3] report substantial efficiency gains over door-to-door baselines, confirming the operational value of flexible pickup locations in many-to-many DRT.

However, the DARPmp of Cortenbach et al. [3] has two important limitations that the present paper addresses. First, the single-sided design leaves dropoff flexibility unexploited: passengers must still be delivered to their

exact destination, which constrains route consolidation on the egress leg. Second, the model treats passenger acceptance as deterministic — once the system assigns a meeting point, the passenger is assumed to comply regardless of walking distance or service quality. In practice, passengers exercise choice, and a model that ignores this risks over-optimistic efficiency estimates. The present paper extends DARPmp to a *bidirectional* design in which both the pickup meeting point $m_r^P \in M_r^P$ and the dropoff meeting point $m_r^D \in M_r^D$ are decision variables, and simulates passenger acceptance with a two-alternative binary logit model.

Fielbaum et al. [12] studied bidirectional walking flexibility in ridepooling, showing that allowing passengers to walk to both pickup and dropoff points reduces vehicle detours and improves system efficiency. Their analysis demonstrates that the efficiency gains from dropoff-side flexibility are comparable in magnitude to those from pickup-side flexibility, providing empirical motivation for the bidirectional design adopted in the present paper. Our work extends this insight to the many-to-many DRT setting with explicit binary logit passenger choice and online re-optimization: unlike Fielbaum et al. [12], who study a static ridepooling context, we address the online assignment problem where meeting points must be selected in real time as requests arrive, and passenger acceptance is modeled probabilistically rather than assumed.

2.3. Passenger Choice in Demand-Responsive Transit

Discrete choice modelling provides the theoretical foundation for representing heterogeneous passenger preferences in transportation systems. Random utility and logit models [13, 14, 15] are widely used in mode choice

analysis and have been applied extensively to transit demand, ridesharing adoption, and on-demand mobility [16]. In this paper, the operational offer mechanism contains two alternatives for each request — accepting the single offered bundle or taking the outside option — so the relevant choice model is binary logit.

The application of discrete choice models to DRT optimization is relatively recent. Most DRT scheduling papers treat passenger acceptance as a binary constraint: a request is either served within its time window or rejected, with no intermediate probability of acceptance. This deterministic treatment ignores the empirical reality that passengers differ in their sensitivity to walking distance, waiting time, in-vehicle travel time, and fare [16]. Heterogeneous preferences are particularly consequential in meeting-point DRT, where the system must select not only *whether* to serve a request but *where* to pick up and drop off the passenger — a choice that directly affects the utility experienced.

The present paper builds on the binary logit framework developed in Jin and Liu [17] for a many-to-one DRT setting with dynamic pricing, extended here to the many-to-many bidirectional meeting-point setting. The core model retains three heterogeneous passenger types and the outside option (passengers may reject the offered bundle). The outside option is critical: it allows the simulation to represent rejection of inconvenient offers, a failure mode absent from deterministic DARP formulations.

2.4. Dynamic Scheduling and Rolling Horizon Methods

Real-world DRT operates in an online environment: requests arrive continuously, vehicles are in motion, and schedules must be updated in near-

real-time. The online DARP [5] and its dynamic variants have been studied extensively, with solution approaches ranging from greedy insertion heuristics to population-based metaheuristics and reinforcement learning [8, 2].

Rolling horizon (receding horizon) methods decompose the infinite-horizon online problem into a sequence of finite-horizon subproblems, each solved to optimality or near-optimality before the horizon advances [18]. Bent and Van Hentenryck [18] established the theoretical basis for rolling horizon approaches in vehicle routing, showing that re-optimization over a sliding window can closely approximate clairvoyant offline solutions when the window length is calibrated to the request arrival rate. This insight motivates the rolling horizon architecture adopted in the present paper.

The Adaptive Large Neighborhood Search (ALNS) of Ropke and Pisinger [19] is the dominant metaheuristic for DARP and related problems, combining destroy-and-repair operators with an adaptive weight mechanism that favors operators performing well on recent iterations. ALNS has been applied to static DARP [6], dynamic ridesharing [9], and, most recently, to online DRT re-optimization.

Wu et al. [4] present a rolling horizon ALNS framework for many-to-many DRT that re-optimises vehicle routes at fixed time intervals as new requests arrive. Their computational experiments on synthetic and real-world instances demonstrate that rolling horizon ALNS achieves high acceptance rates with tractable computation times. However, the formulation of Wu et al. [4] assumes door-to-door service — no meeting points are considered — and treats passenger acceptance as deterministic. The present paper integrates the rolling horizon ALNS architecture of Wu et al. [4] with bidirectional

meeting point assignment and simulated binary logit acceptance.

2.5. Research Gap and Positioning

Table 1 summarizes the positioning of this paper relative to the most closely related prior works. To our knowledge, this is the first simulation framework to combine bidirectional meeting point sets (M_r^P and M_r^D), stochastic passenger acceptance via binary logit with an outside option, rolling horizon re-optimization, and a many-to-many DRT service structure. Fielbaum et al. [12] studied bidirectional walking in static ridepooling, and Alonso-Mora et al. [20] developed dynamic high-capacity trip-vehicle assignment without meeting point choice; our contribution extends these ideas to an online many-to-many DRT setting with simulated acceptance.

Table 1: Positioning relative to prior work

Study	Bidirectional MPs	Passenger Choice	Rolling Horizon	Many-to-M
Cortenbach et al. (2024)	No (single-sided)	No	No	Yes
Wu et al. (2025)	No	No	Yes	Yes
Fielbaum et al. (2021)	Yes (ridepooling)	No	No	Partia
Alonso-Mora et al. (2017)	No	No	No	Yes
This paper	Yes	Yes (binary logit)	Yes	Yes

3. Problem Formulation

3.1. Network and Meeting Points

We model the service area as a directed graph $G = (N, A)$, where N is the set of nodes comprising passenger origins, destinations, and meeting points,

and A is the set of directed arcs. Travel time between any two locations $i, j \in N$ is given by $t(i, j) = d(i, j)/\bar{v}$, where $d(i, j)$ denotes Euclidean distance and $\bar{v}$ is the average vehicle speed.

A finite set $\mathcal{M}$ of pre-defined fixed meeting points (e.g., bus stops, commercial landmarks) is given. For each request r , the candidate pickup set $M_r^P \subseteq \mathcal{M}$ is the subset of meeting points within walking radius ρ^P of the passenger's origin o_r , and the candidate dropoff set $M_r^D \subseteq \mathcal{M}$ is the subset within walking radius ρ^D of the destination δ_r . Formally:

$$M_r^P = \{m \in \mathcal{M} \mid d(o_r, m) \leq \rho^P\}, \quad M_r^D = \{m \in \mathcal{M} \mid d(\delta_r, m) \leq \rho^D\}. \quad (1)$$

Each request $r \in \mathcal{R}$ is characterized by a tuple $(o_r, \delta_r, e_r, l_r, k_r)$, where $[e_r, l_r]$ is the pickup time window and $k_r \in \mathcal{K}$ denotes the passenger type. Requests arrive online over a continuous planning horizon. The fleet $\mathcal{V}$ consists of vehicles with heterogeneous capacities Q_v and maximum shift durations $T_v^{\max}$.

3.2. Service Offer Bundle

For each request r , a *service offer bundle* is a tuple

$$b = (m_r^P, m_r^D, \tau_r, p_r),$$

where:

- $m_r^P \in M_r^P$ is the assigned pickup meeting point,
- $m_r^D \in M_r^D$ is the assigned dropoff meeting point,
- $\tau_r \in [e_r, l_r]$ is the scheduled pickup time at m_r^P ,

- $p_r \geq 0$ is the fare charged to the passenger (fixed in this model).

The set of all feasible bundles for request r is $\mathcal{B}_r = M_r^P \times M_r^D$ (with τ_r and p_r treated as service parameters determined by the system). The service offer bundle b is the central decision object: the system selects one bundle from $\mathcal{B}_r$ or rejects the request.

The bidirectional meeting-point structure distinguishes this model from prior work. In many-to-one DRT, only the pickup side is flexible; here pickup and dropoff meeting points are both decision attributes, creating a richer trade-off between vehicle routing efficiency and passenger walking burden.

3.3. Routing Cost and Welfare Accounting

For offer generation and route construction, the system scores feasible insertions using a weighted sum of operational and passenger service costs:

$$\min \quad \alpha_1 C^{op} + \alpha_2 C^{wait} + \alpha_3 C^{walk} + \alpha_4 C^{IVT}, \quad (2)$$

where:

$$C^{op} = \sum_{v \in \mathcal{V}} c_v \cdot D_v, \quad (3)$$

$$C^{wait} = \sum_{r \in \mathcal{R}^{acc}} (\tau_r - t_r^{req}), \quad (4)$$

$$C^{walk} = \sum_{r \in \mathcal{R}^{acc}} (d(o_r, m_r^P) + d(m_r^D, \delta_r)), \quad (5)$$

$$C^{IVT} = \sum_{r \in \mathcal{R}^{acc}} IVT_{rb}. \quad (6)$$

Here D_v is the total distance driven by vehicle v , c_v is cost per unit distance, and t_r^{req} is the time request r was submitted. The weights $\alpha_1, \dots, \alpha_4 \geq 0$ allow the operator to balance vehicle efficiency against service quality.

For post-hoc policy evaluation, we define a social welfare metric that includes a rejection penalty $\Gamma > 0$:

$$W = \sum_{r \in \mathcal{R}} [z_r \cdot U_{rb^*} - (1 - z_r) \cdot \Gamma]. \quad (7)$$

Γ enters only the ex-post welfare accounting (Section 7.7); it does *not* influence the routing objective (Equation 2), the bundle selection, or the binary logit acceptance probability. The separation ensures that operational decisions are consistent across different policy preferences encoded through Γ .

3.4. Operational Feasibility Conditions

The routing layer enforces the following feasibility conditions for requests whose offers are accepted. These conditions define the simulation's feasible routing state; they are not presented as a complete arc-flow mixed-integer programming formulation.

Vehicle Capacity..

$$q_v(i) \leq Q_v \quad \forall v \in \mathcal{V}, \forall i \in \sigma_v \quad (8)$$

Time Windows..

$$s_{v_r, m_r^P} \geq e_r \quad \forall r \in \mathcal{R}^{\text{acc}} \quad (9)$$

$$s_{v_r, m_r^P} \leq l_r \quad \forall r \in \mathcal{R}^{\text{acc}} \quad (10)$$

In-Vehicle Ride Time..

$$s_{v_r, m_r^D} - s_{v_r, m_r^P} \leq L^{\max} \quad \forall r \in \mathcal{R}^{\text{acc}} \quad (11)$$

Walking Radius..

$$d(o_r, m_r^P) \leq \rho^P \quad \forall r \in \mathcal{R}^{\text{acc}} \quad (12)$$

$$d(\delta_r, m_r^D) \leq \rho^D \quad \forall r \in \mathcal{R}^{\text{acc}} \quad (13)$$

Precedence: Pickup Before Dropoff..

$$\pi_r^P < \pi_r^D \quad \forall r \in \mathcal{R}^{\text{acc}} \quad (14)$$

$$s_{v_r, m_r^P} + t(m_r^P, m_r^D) \leq s_{v_r, m_r^D} \quad \forall r \in \mathcal{R}^{\text{acc}} \quad (15)$$

Route Time Consistency..

$$s_{v, i+1} \geq s_{v, i} + t(\sigma_v(i), \sigma_v(i+1)) \quad \forall v \in \mathcal{V}, \forall i < |\sigma_v| \quad (16)$$

Maximum Route Duration..

$$s_{v, |\sigma_v|} - s_{v, 1} \leq T_v^{\text{max}} \quad \forall v \in \mathcal{V} \quad (17)$$

Domain Constraints..

$$v_r \in \mathcal{V} \quad \forall r \in \mathcal{R}^{\text{acc}} \quad (18)$$

$$m_r^P \in M_r^P \quad \forall r \in \mathcal{R}^{\text{acc}} \quad (19)$$

$$m_r^D \in M_r^D \quad \forall r \in \mathcal{R}^{\text{acc}} \quad (20)$$

Table 2 summarizes the operational feasibility conditions.

3.5. Online Offer and Response Variables

Upon arrival of request r , the system immediately determines the *online offer/routing decision*:

$$\mathbf{a}_r = (m_r^P, m_r^D, v_r, \pi_r^P, \pi_r^D, \tau_r), \quad (21)$$

where $m_r^P \in M_r^P$ and $m_r^D \in M_r^D$ are the assigned meeting points, $v_r \in \mathcal{V}$ is the assigned vehicle, $\pi_r^P < \pi_r^D$ are the insertion positions of the pickup and dropoff nodes in vehicle v_r 's route, and τ_r is the scheduled pickup time implied

Table 2: Summary of operational constraints

Label	Class	Variables Involved
8	Vehicle Capacity	$q_v(i), Q_v, \sigma_v$
9	Time Windows (early)	s_{v_r, m_r^P}, e_r
10	Time Windows (late)	s_{v_r, m_r^P}, l_r
11	In-Vehicle Ride Time	$s_{v_r, m_r^D}, s_{v_r, m_r^P}, L^{\max}$
12	Walking Radius (pickup)	$d(o_r, m_r^P), \rho^P$
13	Walking Radius (dropoff)	$d(\delta_r, m_r^D), \rho^D$
14	Precedence (position)	π_r^P, π_r^D
15	Precedence (time)	$s_{v_r, m_r^P}, s_{v_r, m_r^D}, t(m_r^P, m_r^D)$
16	Route Time Consistency	$s_{v,i}, s_{v,i+1}, t(\sigma_v(i), \sigma_v(i+1))$
17	Maximum Route Duration	$s_{v,1}, s_{v, \sigma_v }, T_v^{\max}$
18	Domain (vehicle)	v_r
19	Domain (pickup point)	m_r^P, M_r^P
20	Domain (dropoff point)	m_r^D, M_r^D

by the insertion. The offer decision must be made online — before future requests are known — and subject to all feasibility constraints above. The binary variable z_r is not optimized by the system; it is the realized passenger response generated after the offer is presented, as defined in Section 4.

4. Passenger Choice Model

4.1. Binary Logit Utility Function

We model passenger response using a two-alternative binary logit model. For request r of passenger type $k \in \mathcal{K}$, the deterministic utility of service

offer bundle $b = (m_r^P, m_r^D, \tau_r, p_r)$ is:

$$U_{rb}^k = \beta_1^k \cdot \text{Walk}_{rb} + \beta_2^k \cdot \text{Wait}_{rb} + \beta_3^k \cdot \text{IVT}_{rb} + \beta_4^k \cdot p_r \quad (22)$$

where:

- $\text{Walk}_{rb} = d(o_r, m_r^P) + d(m_r^D, \delta_r)$ is the total walking distance (pickup walk plus dropoff walk) in meters,
- $\text{Wait}_{rb} = \tau_r - t_r^{\text{req}}$ is the waiting time from request submission to scheduled pickup (minutes),
- $\text{IVT}_{rb} = s_{v_r, m_r^D} - s_{v_r, m_r^P}$ is the in-vehicle travel time from pickup to dropoff meeting point (minutes),
- $p_r \geq 0$ is the fare (CNY),
- $\beta_1^k, \beta_2^k, \beta_3^k \leq 0$ are disutility coefficients for walking, waiting, and in-vehicle time (negative: more time/distance reduces utility),
- $\beta_4^k \leq 0$ is the price disutility coefficient.

The random utility is $U_{rb}^k + \varepsilon_{rb}$ where ε_{rb} is i.i.d. Gumbel(0, 1), yielding the standard random-utility form. The utility function captures the bidirectional walking cost explicitly through Walk_{rb} , which depends on both m_r^P and m_r^D — a key distinction from one-sided meeting-point models.

4.2. Single-Offer Mechanism and Binary Logit Acceptance

The system operates a *single-offer mechanism*: Layer 1 selects the best feasible bundle $b^* \in \mathcal{B}_r$ and presents *only* b^* to the passenger. The passenger

never observes the full set $\mathcal{B}_r$; they choose between accepting b^* or taking the outside option (e.g., private car, walking, trip cancellation).

The utility of the outside option is normalized to zero:

$$U_{r0}^k = 0 \quad (23)$$

Because the passenger's choice set contains exactly two alternatives — b^* and the outside option — the acceptance probability follows a *binary logit* model:

$$P_{\text{accept}}(b^*) = \frac{\exp(U_{rb^*}^k)}{\exp(U_{r0}^k) + \exp(U_{rb^*}^k)} = \frac{\exp(U_{rb^*}^k)}{1 + \exp(U_{rb^*}^k)} \quad (24)$$

The realized acceptance indicator is:

$$z_r \sim \text{Bernoulli}(P_{\text{accept}}(b^*)) \quad (25)$$

This formulation is behaviorally consistent with the single-offer mechanism: the passenger responds to the one bundle they are shown, not to a hypothetical full menu. The standard random-utility interpretation (i.i.d. Gumbel(0, 1) errors) is preserved for this two-alternative setting.

4.3. Passenger Heterogeneity

We distinguish three passenger types $\mathcal{K} = \{\text{price}, \text{time}, \text{walk}\}$ with different sensitivity profiles. Table 3 summarizes the qualitative ordering of coefficients.

Formally, the ordering constraints are:

$$|\beta_4^{\text{price}}| > |\beta_4^{\text{time}}|, \quad |\beta_4^{\text{price}}| > |\beta_4^{\text{walk}}| \quad (26)$$

$$|\beta_2^{\text{time}}| + |\beta_3^{\text{time}}| > |\beta_2^{\text{price}}| + |\beta_3^{\text{price}}| \quad (27)$$

$$|\beta_1^{\text{walk}}| > |\beta_1^{\text{price}}|, \quad |\beta_1^{\text{walk}}| > |\beta_1^{\text{time}}| \quad (28)$$

Table 3: Passenger type coefficient profiles (qualitative ordering)

Type	β_1^k (Walk)	β_2^k (Wait)	β_3^k (IVT)	β_4^k (P)
Price-sensitive	moderate	moderate	moderate	high (most negative)
Time-sensitive	moderate	high (most negative)	high (most negative)	low (most negative)
Walk-sensitive	high (most negative)	moderate	moderate	moderate

Baseline parameter values for simulation:

$$(\beta_1^{\text{price}}, \beta_2^{\text{price}}, \beta_3^{\text{price}}, \beta_4^{\text{price}}) = (-0.005, -0.04, -0.02, -0.15) \quad (29)$$

$$(\beta_1^{\text{time}}, \beta_2^{\text{time}}, \beta_3^{\text{time}}, \beta_4^{\text{time}}) = (-0.005, -0.10, -0.08, -0.03) \quad (30)$$

$$(\beta_1^{\text{walk}}, \beta_2^{\text{walk}}, \beta_3^{\text{walk}}, \beta_4^{\text{walk}}) = (-0.020, -0.04, -0.02, -0.05) \quad (31)$$

Units: β_1 in utility/meter, β_2 and β_3 in utility/minute, β_4 in utility/CNY. The heterogeneous type structure allows the model to capture the distributional equity effects of meeting-point assignment: walk-sensitive passengers (e.g., elderly, mobility-impaired) are disproportionately affected by distant meeting points, a key policy consideration for DRT design in low-density urban areas.¹

¹**Parameter plausibility.** The implied value of time (VOT) for the walk-sensitive type — the most policy-relevant comparator for bidirectional DRT users — is $\text{VOT}_{\text{walk}} = (|\beta_1^{\text{walk}}|/|\beta_4^{\text{walk}}|) \times v_w = (0.020/0.05) \times 80 = 32.0 \text{ CNY/min}$, $\text{VOT}_{\text{wait}} = |\beta_2^{\text{walk}}|/|\beta_4^{\text{walk}}| = 0.04/0.05 = 0.80 \text{ CNY/min}$, and $\text{VOT}_{\text{IVT}} = 0.02/0.05 = 0.40 \text{ CNY/min}$ (walking speed $v_w = 80 \text{ m/min}$ assumed). The wait and IVT VOT values (0.80 and 0.40 CNY/min) fall within the Chinese urban benchmark range of 0.8–1.2 CNY/min (waiting) and 0.2–0.4 CNY/min (IVT) reported by Shao et al. [21] for Beijing commuters and Li et al. [22] for DRT users. The walk VOT (32.0 CNY/min) exceeds the literature range

Worked utility example. Consider a walk-sensitive passenger ($k = \text{walk}$) with $\beta_1^{\text{walk}} = -0.020$ utils/m, $\beta_2^{\text{walk}} = -0.04$ utils/min, $\beta_3^{\text{walk}} = -0.02$ utils/min, and $\beta_4^{\text{walk}} = -0.05$ utils/CNY. Suppose the system offers a bundle with walking distance $d_{\text{walk}} = 300$ m, waiting time $t_{\text{wait}} = 5$ min, in-vehicle time $t_{\text{ivt}} = 20$ min, and fare $p = 8$ CNY. The deterministic utility is:

$$\begin{aligned} U &= \beta_1 \cdot d_{\text{walk}} + \beta_2 \cdot t_{\text{wait}} + \beta_3 \cdot t_{\text{ivt}} + \beta_4 \cdot p \\ &= (-0.020)(300) + (-0.04)(5) + (-0.02)(20) + (-0.05)(8) \\ &= -6.0 - 0.2 - 0.4 - 0.4 = -7.0 \text{ utils.} \end{aligned}$$

With the outside option utility normalized to zero ($U_{r0}^k = 0$, Equation 23), the acceptance probability is:

$$P_{\text{accept}} = \frac{e^{-7.0}}{e^{-7.0} + e^0} = \frac{0.000912}{0.000912 + 1} \approx 0.00091. \quad (32)$$

This near-zero acceptance probability (0.091%) illustrates why the offer quality generated by the online algorithm is critical: under the $U_0 = 0$ normalization, any bundle with strongly negative utility will be rejected with very high probability. The system must therefore offer bundles with substantially higher utility (e.g., by reducing walking distance, lowering wait times, or offering price discounts) to achieve meaningful acceptance rates. Reducing

(0.4–0.6 CNY/min); this reflects the strong walking disutility assumed for the walk-sensitive type ($|\beta_1^{\text{walk}}| = 0.020$ utils/meter) combined with low price sensitivity ($|\beta_4^{\text{walk}}| = 0.05$ utils/CNY). The resulting ratio ($0.020/0.05 \times 80 = 32.0$ CNY/min) is an upper bound; calibration against revealed-preference data from Chinese DRT users would likely yield lower walking disutility coefficients. This limitation is discussed in Section 8 (VOT interpretation) and Section 9. type’s defining characteristic. Full VOT mapping across all three passenger types is reported in Table 14 (Section 8).

walking to 100 m would raise U to -3.0 , yielding $P_{\text{accept}} \approx 0.047$ (4.7% acceptance), illustrating the sensitivity of demand to meeting point placement.

5. Three-Layer Coupled Model

The many-to-many DRT system with bidirectional meeting points operates through three sequentially coupled layers, executed online upon each request arrival and periodically re-optimized via a rolling horizon mechanism. Figure 2 illustrates the architecture. The three layers are: (1) service offer generation, (2) passenger response, and (3) dynamic dispatch and re-optimization.

5.1. Layer 1: Service Offer Generation

Upon arrival of request r at time t_r^{req} , the system generates a set of candidate service offer bundles $\hat{\mathcal{B}}_r \subseteq \mathcal{B}_r$ as follows:

1. **Candidate pickup set:** Compute $M_r^P = \{m \in \mathcal{M} \mid d(o_r, m) \leq \rho^P\}$. Retain the top- k^{top} candidates ranked by proximity to o_r .
2. **Candidate dropoff set:** Compute $M_r^D = \{m \in \mathcal{M} \mid d(\delta_r, m) \leq \rho^D\}$. Retain the top- k^{top} candidates ranked by proximity to δ_r .
3. **Bundle enumeration:** For each pair $(m^P, m^D) \in M_r^P \times M_r^D$, evaluate feasibility against constraints 8–16 for each vehicle $v \in \mathcal{V}$ and each insertion position pair (π^P, π^D) .
4. **Bundle selection:** Select the bundle $b^* = \arg \min_{b \in \hat{\mathcal{B}}_r} \Delta C_{\text{routing}}(b, \{\sigma_v\})$ that minimizes the deterministic incremental routing cost $\alpha_1 \Delta C^{\text{op}} + \alpha_2 \Delta C^{\text{wait}} + \alpha_3 \Delta C^{\text{walk}} + \alpha_4 \Delta C^{\text{IVT}}$ (see Section 6 for implementation details).

The output of Layer 1 is the online offer/routing decision $\mathbf{a}_r$ (Equation 21) and the service offer bundle b^* presented to the passenger.

5.2. Layer 2: Passenger Response

Given the offered bundle b^* , passenger r of type k accepts with probability $P_{\text{accept}}(b^*)$ (Equation 24) and rejects with probability $1 - P_{\text{accept}}(b^*)$. The realized acceptance indicator is:

$$z_r \sim \text{Bernoulli}(P_{\text{accept}}(b^*)) \quad (33)$$

If $z_r = 1$, the vehicle route σ_{v_r} is updated by inserting the pickup node m_r^P at position π_r^P and the dropoff node m_r^D at position π_r^D . If $z_r = 0$, the offer is rejected and vehicle routes remain unchanged.

The coupling between Layer 1 and Layer 2 follows a *route-then-sample* design (Section 6): Layer 1 selects b^* by minimizing deterministic incremental routing cost; the binary logit acceptance probability $P_{\text{accept}}(b^*)$ is computed after bundle selection and used only for the Bernoulli acceptance sampling in Layer 2. This decoupling avoids the perverse incentive where multiplying P_{accept} into the routing objective would favor low-acceptance (poor service) offers. Integrating P_{accept} into a welfare-consistent anticipatory objective remains a direction for future work.

5.3. Layer 3: Dynamic Dispatch and Rolling Horizon Re-optimization

Vehicle dispatch follows the committed route schedules $\{\sigma_v, \mathbf{s}_v\}_{v \in \mathcal{V}}$. Every Δ minutes, a rolling horizon re-optimization is triggered over the active request window $[t^{\text{now}}, t^{\text{now}} + H]$:

1. **Active set:** Identify $\mathcal{R}^{\text{act}}(t^{\text{now}})$ — all accepted requests not yet completed at time t^{now} .
2. **Re-optimization:** Apply ALNS (Adaptive Large Neighborhood Search) to reassign and reorder requests in $\mathcal{R}^{\text{act}}(t^{\text{now}})$ across vehicles, subject to constraints 8–16.
3. **Commitment:** Commit the updated routes for the next Δ minutes; nodes already served are locked and cannot be reassigned.

The rolling horizon creates a feedback loop from Layer 3 back to Layer 1: updated vehicle schedules after re-optimization change the feasibility and cost of future bundle insertions, influencing the next Layer 1 evaluation.

$$\text{Re-optimize at times } t^{\text{now}} = 0, \Delta, 2\Delta, 3\Delta, \dots \quad (34)$$

Typical parameter values: $H = 30$ minutes (horizon window), $\Delta = 5$ minutes (re-optimization frequency). These are configurable and subject to sensitivity analysis in the sensitivity analysis (Section 7).

5.4. Layer Coupling Summary

Table 4 summarizes the information flows between layers.

Table 4: Three-layer model coupling structure

Layer	Input	Output	Coupling
1: Service Generation	Request r , routes $\{\sigma_v\}$	Bundle b^* , decision $\mathbf{a}_r$	Feeds Layer 2
2: Passenger Response	Bundle b^* , type k	Acceptance z_r	Updates routes if $z_r =$
3: Rolling Horizon	Active set $\mathcal{R}^{\text{act}}$	Updated routes $\{\sigma_v\}$	Feeds back to Layer 1

Table 5: Timing and decision sequence in the rolling horizon three-layer model. Each row represents one event within a single planning cycle. Re-optimization triggers fire at $t^{\text{now}} = 0, \Delta, 2\Delta, \dots$ (Equation 34).

Event	Layer	Variables / out-comes	Information Flow	Bernoulli?
Request r arrives at t_r^{req}	1	$m_r^P, m_r^D, v_r, \pi_r^P, \pi_r^D$	Routes $\{\sigma_v\}$ in; bundle b^* out	No
Bundle b^* pre-sented to passenger	2	z_r	b^* in; acceptance z_r out	Yes: $z_r \sim \text{Bern}(P_{\text{acc}})$
If $z_r = 1$: route σ_{v_r} updated	3	(route update)	z_r, b^* in; updated σ_{v_r} out	No
Re-opt trigger at $t = k\Delta$	3	$\{\sigma_v\}$ (all vehicles)	$\mathcal{R}^{\text{act}}$ in; optimised $\{\sigma_v\}$ out	No
Updated routes fed back for next re-quest	1	—	Updated $\{\sigma_v\}$ avail-able to Layer 1	No

Table 5 details the event sequence and information flows within a single planning cycle; in particular, it shows that Bernoulli sampling ($z_r \sim \text{Bernoulli}(P_{\text{accept}}(b^*))$) occurs exclusively in Layer 2, after bundle b^* has been selected by Layer 1.

The three-layer architecture captures the essential feedback structure of online DRT operations: routing decisions shape the offers available to passengers, passenger choices determine realized demand, and periodic re-optimization adapts routes to the evolving state. The notation table (Table A.15) in the appendix provides a complete reference for all symbols used in this paper.

6. Solution Methodology

The problem defined in Section 3 is NP-hard (see Section 6.3). We develop two complementary solution approaches: an ex-post MILP routing diagnostic solved by Gurobi for small instances, and a rolling horizon Adaptive Large Neighborhood Search (ALNS) heuristic for real-time operation at scale.

6.1. Ex-Post Routing Diagnostic: MILP Formulation

For small instances, we formulate the static batch routing problem as a Mixed-Integer Linear Program (MILP) to compute the optimal routing cost for a given accepted set. The MILP serves as an *ex-post routing diagnostic*, not a validation tool for the complete simulation framework, which includes offer generation, routing, and stochastic passenger response.

Decision Variables.. The MILP uses binary assignment variables $x_{r,m^P,m^D,v} \in \{0,1\}$, equal to 1 if request r is assigned to meeting point pair $(m^P, m^D) \in M_r^P \times M_r^D$ and served by vehicle $v \in \mathcal{V}$. Continuous variables $s_{v,i} \geq 0$ represent the scheduled service time at node i in vehicle v 's route. The MILP is solved only for the fixed realized set $\mathcal{R}^{\text{acc}}$; it does not optimize the passenger response variable z_r .

Objective.. The MILP minimizes the same weighted cost (Equation 2), with each cost component linearized in terms of $x_{r,m^P,m^D,v}$ and $s_{v,i}$.

Constraints.. The formulation enforces the operational feasibility conditions (8)–(20). Capacity constraint (8) is linearized using standard load-tracking variables. Time-window constraints (9)–(10), ride-time constraint (11), and

route time consistency (16) are expressed as linear inequalities in $s_{v,i}$. Walking radius constraints (12)–(13) are enforced implicitly by restricting the domain of $x_{r,m^P,m^D,v}$ to feasible meeting point pairs.

Solver and Scope. The MILP is solved using Gurobi 10 with a 300-second time limit; Gurobi reports the optimality gap for instances where the time limit binds.

Ex-post diagnostic role. The MILP serves as an *ex-post* routing diagnostic, not a real-time solver. Given the realized accepted set $\mathcal{R}^{\text{acc}}$ produced by a completed ALNS run — i.e., the set of requests for which the sampled acceptance indicator $z_r = 1$ — the MILP solves the optimal vehicle routing for that *fixed* accepted set. Formally, the MILP treats z_r as a known constant (equal to 1 for $r \in \mathcal{R}^{\text{acc}}$ and 0 otherwise) and optimizes only over the assignment variables $x_{r,m^P,m^D,v}$ and schedule variables $s_{v,i}$. It does *not* re-optimize the acceptance decisions: the stochastic passenger responses are realized outcomes of the ALNS run, not MILP decision variables. The resulting MILP objective value is therefore a lower bound on the routing cost achievable for the same accepted set, and the diagnostic gap $\delta = (C^{\text{ALNS}} - C^{\text{MILP}})/C^{\text{MILP}} \times 100\%$ measures how close the ALNS heuristic comes to the ex-post routing optimum on this fixed, realized set only. Instances with up to 30–50 passengers and 5–8 vehicles are tractable within the time limit.

6.2. Heuristic Algorithm: Rolling Horizon ALNS

For real-time operation, we develop a rolling horizon ALNS heuristic that processes arriving requests online and periodically re-optimizes committed

routes. The algorithm follows the three-layer architecture of Section 5.

Online Insertion Mechanism.. The ALNS heuristic uses a *route-then-sample* mechanism that decouples stochastic passenger acceptance from deterministic routing optimization. At each request arrival, the system performs the following steps:

1. **Bundle selection (Layer 1):** The system selects the best feasible bundle b^* by minimizing the *deterministic* incremental routing cost $\Delta C_{\text{routing}}(b, \{\sigma_v\})$, evaluated as the weighted sum $\alpha_1 \Delta C^{\text{op}} + \alpha_2 \Delta C^{\text{wait}} + \alpha_3 \Delta C^{\text{walk}} + \alpha_4 \Delta C^{\text{IVT}}$ of inserting bundle b into the current vehicle schedules $\{\sigma_v\}$. The acceptance probability $P_{\text{accept}}(b^*)$ is *not* part of the bundle-selection objective.
2. **Acceptance sampling (Layer 2):** After bundle selection, the binary logit acceptance probability $P_{\text{accept}}(b^*) = \exp(U_{rb^*}^k) / (1 + \exp(U_{rb^*}^k))$ (Equation 24) is computed, and the passenger’s response is sampled as $z_r \sim \text{Bernoulli}(P_{\text{accept}}(b^*))$. If $z_r = 1$, the bundle is inserted into the vehicle route.

Rationale for route-then-sample.. This design separates the stochastic acceptance decision from the deterministic cost minimization. An alternative approach that integrates P_{accept} directly into the routing objective as $\mathbb{E}[\Delta C] = P_{\text{accept}}(b) \times \Delta C_{\text{routing}}(b)$ would create a perverse incentive: bundles with lower acceptance probability (worse service quality) would appear to have lower expected routing cost, driving the system toward offering unattractive bundles. The route-then-sample approach avoids this pathology while preserving the structural benefit of passenger self-selection: passengers

who would face high walking or waiting disutility reject the service, and the system concentrates vehicle resources on spatially compatible requests. Integrating P_{accept} into the routing objective in a welfare-consistent manner remains a direction for future work.

Rejection penalty (post-hoc only). The rejection penalty Γ enters only the social welfare metric $W = \sum_r [z_r U_{rb^*} - (1 - z_r)\Gamma]$ (Section 7.7), which is a post-hoc evaluation tool. Γ does *not* enter the bundle selection, the routing objective, or the binary logit acceptance probability. The acceptance probability $P_{\text{accept}}(b^*)$ is invariant to Γ . This separation ensures that the operational decisions are consistent across different policy preferences encoded through Γ .

6.2.1. Candidate Generation

Upon arrival of request r , the system generates candidate meeting point sets M_r^P and M_r^D by filtering $\mathcal{M}$ against walking radii ρ^P and ρ^D (constraints 12–13), then retaining the top- k^{top} candidates ranked by proximity to o_r and δ_r respectively. This yields $|M_r^P| \leq k^{\text{top}}$ pickup candidates and $|M_r^D| \leq k^{\text{top}}$ dropoff candidates. The filtering step runs in $O(|\mathcal{M}| \log |\mathcal{M}|)$ per request using a pre-sorted distance index.

6.2.2. Online Insertion Evaluation

For each bundle $(m^P, m^D) \in M_r^P \times M_r^D$, the algorithm enumerates all vehicles $v \in \mathcal{V}$ and all insertion position pairs (π^P, π^D) with $\pi^D > \pi^P$ (constraint 14). Feasibility against constraints (8), (9)–(10), (11), and (16) is checked in $O(1)$ per position pair using forward and backward schedule arrays maintained for each vehicle route. The best feasible insertion is selected

by minimum incremental cost ΔC to the objective (2). After selecting b^* , the binary logit acceptance probability $P_{\text{accept}}(b^*) = \exp(U_{rb^*}^k)/(1+\exp(U_{rb^*}^k))$ (Equation 24) is computed and the passenger’s response is sampled as $z_r \sim \text{Bernoulli}(P_{\text{accept}}(b^*))$.

6.2.3. ALNS Destroy and Repair Operators

Every Δ minutes, the rolling horizon re-optimization applies ALNS [19] to the active request set $\mathcal{R}^{\text{act}}(t^{\text{now}})$. The algorithm uses five operators:

1. **Request reassignment:** Remove k randomly selected requests from their current vehicle routes and reinsert them greedily into the best feasible positions across all vehicles.
2. **Pickup/dropoff pair swap:** Exchange the meeting point pair (m_r^P, m_r^D) assignments between two requests r and r' , updating insertion positions accordingly.
3. **Insertion position exchange:** Swap the route positions of two service nodes on the same vehicle, subject to precedence and feasibility.
4. **Route segment reconstruction:** Remove a contiguous route segment from a vehicle, then reinsert the affected requests optimally across all vehicles.
5. **Worst removal:** Remove the k requests with the highest marginal cost contribution to the objective, then reinsert greedily.

Operator selection follows the standard ALNS roulette-wheel mechanism [19]: each operator maintains a score updated by its historical improvement contribution, and operators are selected with probability proportional to their

score. Meeting point reassignment is integrated into the repair step of operators 1, 4, and 5: when a request is reinserted, the algorithm re-evaluates all (m^P, m^D) pairs in $M_r^P \times M_r^D$ and selects the pair that minimizes insertion cost.

6.2.4. Rolling Horizon Loop

The rolling horizon framework [18] triggers ALNS re-optimization at times $t^{\text{now}} = 0, \Delta, 2\Delta, \dots$ (Equation 34). At each trigger, the algorithm operates on the active set $\mathcal{R}^{\text{act}}(t^{\text{now}})$ over the window $[t^{\text{now}}, t^{\text{now}} + H]$ with $H = 30$ minutes and $\Delta = 5$ minutes. Committed nodes (already served or within the commitment horizon) are locked and cannot be reassigned.

Commitment assumption.. Once a service offer bundle $b_r^* = (m_r^P, m_r^D, \tau_r, p_r)$ is communicated to passenger r and accepted ($z_r = 1$), the assigned meeting points and scheduled pickup time are *committed*: they are not subject to re-optimization in subsequent rolling horizon cycles. This commitment is operationally necessary — passengers who have begun walking to a meeting point cannot be redirected — and is enforced by locking the corresponding nodes in the route graph before each ALNS re-optimization trigger. Formally, a node $i \in \sigma_v$ is committed if its scheduled service time satisfies $\tau_i \leq t^{\text{now}} + \delta_{\text{commit}}$, where $\delta_{\text{commit}} = 5$ minutes is the commitment horizon (equal to the re-optimization interval Δ). Uncommitted accepted nodes may be re-sequenced within the vehicle route to improve efficiency, but their assigned meeting points remain fixed.

The ALNS runs for 50 iterations per re-optimization trigger, with each iteration applying one destroy–repair operator pair. After re-optimization,

updated routes are committed for the next Δ minutes and fed back to Layer 1 for subsequent request evaluations.

6.3. Computational Complexity

NP-Hardness. The many-to-many DRT problem with bidirectional meeting points is NP-hard. This follows by reduction from the Dial-a-Ride Problem (DARP) [1]: for each request r , set the candidate pickup set to a singleton containing only the passenger’s origin ($M_r^P = \{o_r\}$) and the candidate dropoff set to a singleton containing only the destination ($M_r^D = \{\delta_r\}$), with ρ^P and ρ^D set sufficiently large to include these singletons. The problem then reduces to standard DARP (door-to-door with no meeting point choice), which is NP-hard. The bidirectional meeting-point extension strictly generalizes this case by allowing $|M_r^P|, |M_r^D| \geq 1$, adding the joint optimization of m_r^P and m_r^D , so the problem is at least as hard.

Heuristic Complexity. The per-request candidate generation step runs in $O(|\mathcal{M}| \log |\mathcal{M}|)$. Online insertion evaluation for a single request costs $O(|\mathcal{V}| \cdot |\sigma_v|^2 \cdot (k^{\text{top}})^2)$, where $|\sigma_v|$ is the average route length. Each ALNS iteration costs $O(n \cdot |\mathcal{V}| \cdot |\sigma_v|)$ where $n = |\mathcal{R}^{\text{act}}|$. The rolling horizon bounds the active set size: at any trigger, at most $H/\bar{\lambda}^{-1}$ requests are active, where $\bar{\lambda}$ is the average arrival rate. This keeps per-step complexity manageable even as the total number of requests grows.

Practical Performance. On the 200-request, 15-vehicle simulation instances, the average decision time per request is below 1 second, satisfying the real-time requirement of online DRT operation. The MILP ex-post routing diagnostic (Section 6.1) evaluates how close the ALNS routing comes to the

optimal offline routing for a fixed accepted set on small instances (gaps of 99–170% for $n = 20$ –30; see Table 8). These gaps reflect the inherent difficulty of online insertion with fixed meeting points relative to optimal ex-post routing, not an assessment of overall solution quality, since the MILP does not model stochastic passenger response.

7. Numerical Experiments

This section evaluates the proposed bidirectional meeting point model through controlled numerical experiments. We compare six model variants across two scenarios, conduct sensitivity analyses on the key deployment parameters, and examine service equity across passenger types.

7.1. Experimental Setup

Scenarios. We use two complementary scenarios. The *synthetic scenario* places requests uniformly at random on a $20\text{ km} \times 20\text{ km}$ grid. Demand ranges from 100 to 500 requests per horizon; fleet size ranges from 10 to 30 vehicles. Vehicle capacity is 8 passengers and the maximum ride time is 45 minutes. This scenario allows controlled parameter sweeps with full reproducibility (seeds 42, 43, 44). Results are averaged over three random seeds (42, 43, 44), consistent with standard practice in DRT simulation [4]; standard deviations are reported in Table 7.

The *Beijing-inspired synthetic grid scenario* is calibrated to a Beijing suburban district: a $15\text{ km} \times 15\text{ km}$ service zone with 81 meeting points arranged on a complete 9×9 grid at 1875 m spacing, 200 requests during the morning peak (7–9 am), and 15 vehicles. This Beijing-inspired synthetic grid

scenario tests whether the efficiency gains observed in the synthetic setting extend to a denser meeting point configuration.

Choice model parameters.. The binary logit choice model (Section 4) uses the type-specific four-attribute utility coefficients defined in Equations 29–31. The outside option utility is normalized to zero ($U_{r_0}^k = 0$, Equation 23). Three passenger types are defined — price-sensitive, time-sensitive, and walk-sensitive — each comprising approximately one-third of the demand. The per-type beta values and the route-then-sample implementation are documented in Sections 4 and 6.

Model variants.. Six variants are evaluated, differing across three dimensions: meeting point configuration (none / single-sided / bidirectional), response mechanism (none / binary logit sampling / deterministic diagnostic), and re-optimization (rolling horizon ALNS / static greedy). Table 6 summarizes the design of each variant.

Table 6: Model variant definitions across three design dimensions

Variant	Meeting Points	Choice Mechanism	Re-optimization
DoorToDoor	None (O-D direct)	All feasible accepted	Rolling horizon
SingleSidedPickup	Pickup only	All feasible accepted	Rolling horizon
BidirectionalNoChoice	Bidirectional	All feasible accepted	Rolling horizon
FullModel	Bidirectional	Binary logit sampling	Rolling horizon
AblationNoChoice	Bidirectional	Deterministic insertion diagnostic	Rolling horizon
AblationNoRollingHorizon	Bidirectional	Binary logit sampling	Static greedy

DoorToDoor serves requests directly, without meeting points or binary

logit acceptance. **SingleSidedPickup** assigns pickup meeting points only. **BidirectionalNoChoice** assigns bidirectional meeting points deterministically (all feasible requests accepted). **FullModel** is the proposed simulation with bidirectional meeting points and binary logit passenger acceptance (route-then-sample mechanism, Section 6). **AblationNoChoice** is a diagnostic variant that removes the stochastic passenger-response layer and uses deterministic greedy insertion only; its reported served share is therefore the share inserted by that deterministic heuristic, not a behavioral acceptance rate. **AblationNoRollingHorizon** disables the rolling horizon, using static greedy insertion instead.

Performance metrics.. Nine metrics are recorded: acceptance rate, vehicle-km (vkm), average and 95th-percentile waiting time, average walking distance, average in-vehicle time (IVT), detour ratio, fairness index (Gini coefficient of per-type acceptance rates), and CPU time per horizon. We additionally report *social welfare* $W = \sum_r [z_r \cdot U_{rb^*} - (1 - z_r) \cdot \Gamma]$, where $z_r \in \{0, 1\}$ is the acceptance indicator, U_{rb^*} is the accepted passenger’s utility under the best offer, and $\Gamma \geq 0$ is a rejection penalty that governs the coverage–efficiency tradeoff (see Section 7.7). All results are averaged across three seeds.

7.2. Main Results: Baseline Comparison

Table 7 reports mean performance across the six variants for the synthetic scenario with 200 requests and 15 vehicles. Vkm/trip values use accepted trip count as the denominator ($n \cdot \hat{\alpha}$). **Note:** the binary logit acceptance model (Section 4) applies only to FullModel and AblationNoRollingHorizon.

DoorToDoor, SingleSidedPickup, and BidirectionalNoChoice accept all feasible requests; DoorToDoor’s acceptance (61.0%) reflects infeasibility rejections, not binary logit sampling. The primary efficiency claim uses the unconstrained baseline comparison in Table 7. The matched-coverage comparison in Table 9 is reported as a separate exploratory diagnostic for coverage confounding, not as the primary headline estimate. The Γ and weight-sensitivity tables are separate diagnostic runs and are not used to compute the 29.1% unconstrained figure.

Table 7: Performance comparison across model variants (synthetic scenario, 200 requests, 15 vehicles, mean $\pm$ std)

Variant	Serv./Ins.	vkm	vkm/trip	Avg Wait (s)	Avg Wa
DoorToDoor	0.610 ± 0.118	2603.7 ± 1666.4	21.3	5003.5	0.0
SingleSidedPickup	0.440 ± 0.275	1199.4 ± 970.6	13.6	3847.1	1023
BidirectionalNoChoice	0.533 ± 0.193	1721.6 ± 1149.0	16.1	4626.7	1652
FullModel	0.183 ± 0.018	552.0 ± 343.3	15.1	454.2	109
AblationNoChoice	0.135 ± 0.245	159.3 ± 290.1	5.9	118.1	25
AblationNoRollingHorizon	0.206 ± 0.062	668.5 ± 373.9	16.2	3114.5	1803

Note: “Serv./Ins.” denotes served share for behavioral variants and inserted share for the deterministic insertion diagnostic AblationNoChoice; the latter is not a behavioral acceptance rate.

7.3. MILP vs. ALNS Optimality Gap

To diagnose routing sub-optimality, we compare ALNS against the MILP ex-post routing diagnostic on small instances where the MILP is tractable. For each instance, ALNS is run first to obtain the realized accepted set $\mathcal{R}^{\text{acc}}$;

the MILP then solves the optimal vehicle routing for that *fixed* accepted set (see Section 6.1). The optimality gap is $\delta = (C^{\text{ALNS}} - C^{\text{MILP}})/C^{\text{MILP}} \times 100\%$.

Table 8: MILP vs. ALNS optimality gap on small instances (mean $\pm$ std across 3 seeds)

n	ALNS vkm	MILP vkm	Gap (%)	MILP status	Solve time (s)
20	54.1	21.7	169.6 ± 46.8	optimal	< 0.001
30	65.6	33.3	98.8 ± 40.3	optimal	< 0.001

The ALNS heuristic solutions are 170% above the MILP ex-post optimum on $n = 20$ instances and 99% above on $n = 30$ instances. These large gaps reflect the very small accepted-set sizes (4–7 passengers per instance, MILP solve time < 0.001 s); the MILP solves an over-simplified routing problem on a known fixed set, not a meaningful validation of ALNS performance. We retain this comparison only as a minimal sanity check that the ALNS solution is feasible; it does not validate heuristic quality at operational scales.

Efficiency and coverage confounding.. A natural concern is whether FullModel’s lower vehicle-km is endogenous: fewer served trips mechanically reduce total driving distance. The primary comparison in Table 7 is therefore the unconstrained per-served-trip comparison: FullModel serves 18.3% of requests at 15.1 vkm/trip, while DoorToDoor serves 61.0% at 21.3 vkm/trip, a 29.1% lower vkm/trip for FullModel.

To probe coverage confounding, we also implement a separate *DoorToDoor (capped)* diagnostic that enforces an endogenous acceptance cap: once the served share reaches FullModel’s mean of 18.3%, all subsequent requests are rejected without route insertion, and the ALNS continues re-optimising routes

for already-accepted passengers. This ensures DoorToDoor is re-routed at approximately the same coverage level, not merely post-hoc subsampled. In this diagnostic run (Table 9), FullModel achieves 11.1 vkm/trip compared to 17.1 vkm/trip for DoorToDoor (capped), a 35.0% difference. Because this matched-coverage value comes from a separate diagnostic run rather than the same realization as Table 7, we treat it as exploratory evidence that the efficiency pattern is not solely driven by coverage differences, while retaining the 29.1% unconstrained comparison as the primary headline result.

Table 9: Exploratory matched-coverage diagnostic: FullModel vs. DoorToDoor (capped) at approximately equal served share, mean across 3 seeds. DoorToDoor (capped) achieves a share close to FullModel’s, with minor overshoot due to discrete request granularity.

Variant	Served share	vkm/trip
FullModel	~18%	11.1
DoorToDoor (capped)	~18%	17.1

Table 10: Provenance of the main efficiency numbers.

Result source	Role in paper		Served share		Reported vkm/trip
Table 7	Primary	unconstrained comparison	FullModel	18.3%,	15.1 vs. 21.3
			DoorToDoor	61.0%	
Table 9	Exploratory	matched-coverage diagnostic	Both	approximately 18%	11.1 vs. 17.1

Role of passenger choice. Comparing FullModel to AblationNoChoice provides a diagnostic view of the binary logit response layer. AblationNoChoice

has a lower inserted share (13.5%) and vehicle-km (159.3) than FullModel because it uses a deterministic greedy insertion routine rather than the route-then-sample simulation. This variant should therefore be interpreted as an algorithmic diagnostic, not as a behavioral acceptance variant. The FullModel’s selective acceptance concentrates service on spatially compatible requests, improving per-trip routing efficiency.

Role of rolling horizon.. AblationNoRollingHorizon reveals the cost of static assignment: average waiting time rises to 3114.5 s (versus 454.2 s for FullModel) — a 6.9-fold increase — and the detour ratio reaches 6.14. Without periodic re-optimization, early-assigned vehicles accumulate detours that cannot be corrected, leading to cascading delays. The 21.1% increase in vehicle-km (668.5 vs. 552.0) further confirms that rolling horizon re-optimization is essential for operational efficiency.

Single-sided vs. bidirectional.. SingleSidedPickup reduces vehicle-km by 53.9% relative to DoorToDoor (1199.4 vs. 2603.7) but imposes a 1025.3 m average walk at dropoff. BidirectionalNoChoice achieves a higher acceptance rate (53.3%) but at higher vehicle-km (1721.6) and longer waits (4626.7 s) than FullModel, because without the choice model the optimizer cannot anticipate which meeting point assignments passengers will accept.

7.4. Beijing-Inspired Synthetic Grid Scenario Results

Table 11 reports results for the Beijing-inspired synthetic grid scenario (15 km $\times$ 15 km, 9 $\times$ 9 meeting point grid at 1875 m spacing, 200 requests during morning peak 7–9 am, 15 vehicles). The pattern of results is consistent with the synthetic scenario: FullModel achieves substantially lower

vehicle-km (261.7) than DoorToDoor (1353.9) at lower acceptance (31.3% vs. 77.5%), with the efficiency gain arising from spatial consolidation of accepted requests. The Beijing grid’s denser meeting point spacing (1875 m vs. 2000 m) yields higher absolute acceptance rates for all variants and lower per-trip vehicle-km (FullModel 4.2 vs. 15.1 vkm/trip in the synthetic baseline), consistent with the walking tolerance sensitivity findings in Section 7.

Table 11: Beijing-inspired synthetic grid scenario results (200 requests, 15 vehicles, 9×9 MP grid, 1875 m spacing, morning peak 7–9 am, mean across up to 3 seeds)

Variant	Serv./Ins.	vkm	vkm/trip	Avg Wait (s)	Avg Walk (m)	CPU
DoorToDoor	0.775	1353.9	8.7	7225.2	0.0	10
DoorToDoor (capped)	0.235	480.3	10.2	6172.5	0.0	0
SingleSidedPickup	0.780	1227.6	7.9	7691.8	1252.0	82
BidirectionalNoChoice	0.813	1221.1	7.5	8081.7	1700.2	52
FullModel	0.313	261.7	4.2	350.0	40.0	4
AblationNoChoice	0.765	554.5	3.6	291.1	79.7	11
AblationNoRollingHorizon	0.360	506.8	7.0	6089.0	1639.1	4

7.5. Sensitivity Analysis

Walking tolerance ρ . We sweep $\rho \in \{200, 400, 600, 800, 1000\}$ m with all other parameters fixed at baseline (200 requests, 15 vehicles). FullModel acceptance rate is **0.0%** at $\rho \leq 400$ m: the binary logit utility function penalizes walking so heavily that no meeting-point offer is accepted when the required walk exceeds 400 m (under the current parameterization). Acceptance rises monotonically, reaching **9.0%** at $\rho = 1000$ m. DoorToDoor holds

steady at approximately 58% regardless of ρ , since it imposes no walking requirement and does not use the binary logit acceptance model.

In our simulation setting, acceptance becomes non-zero only at $\rho \geq 1000\text{m}$. Below this threshold, the system degenerates to near-zero acceptance under the current utility parameters. This is a joint consequence of the meeting point spacing (2km grid), the binary logit parameters, and the $U_0 = 0$ normalization; operators should re-evaluate this threshold using locally calibrated parameters.

Fleet size.. We sweep $n_{\text{vehicles}} \in \{5, 10, 15, 20, 25, 30\}$ with $n_{\text{requests}} = 200$ fixed. FullModel acceptance peaks at **24.0%** with $n_{\text{vehicles}} = 30$ (15 vehicles per 100 requests). The marginal gain in acceptance rate diminishes beyond 20 vehicles: adding vehicles 21–30 yields only 2.1 percentage points of additional acceptance, compared to 8.3 pp for vehicles 11–20. This diminishing returns pattern indicates that vehicle scarcity is the binding constraint at low fleet sizes, while at higher fleet sizes the binary logit outside option becomes the binding constraint.

At $n_{\text{vehicles}} = 15$ (baseline), DoorToDoor acceptance exceeds FullModel by 43 percentage points (61.0% vs. 18.3%), confirming that door-to-door service is more robust under vehicle scarcity because it does not impose walking penalties. Operators should therefore ensure adequate fleet supply before deploying bidirectional meeting points.

7.6. Equity Analysis

Table 7 reports aggregate acceptance rates, but these mask heterogeneity across passenger types. We disaggregate FullModel acceptance by type using

the equity analysis described in Section 7.

Per-type acceptance rates. Price-sensitive passengers achieve the highest acceptance rate at **25.4%**, followed by walk-sensitive passengers at **20.2%**, and time-sensitive passengers at **14.4%**. The 11 percentage point gap between price-sensitive and time-sensitive passengers reflects a systematic disadvantage: time-sensitive passengers face longer in-vehicle travel times (avg IVT 1109.6s) and higher waiting times (avg 603.2s) relative to other types, making meeting-point offers less attractive under their utility function.

Gini coefficient. The Gini coefficient of per-type acceptance rates is **0.1216**, indicating moderate inequality. Approximately 12% of the total acceptance rate variation is attributable to passenger type rather than random demand variation. While this level of inequality is lower than that observed in DoorToDoor (Gini = 0.5385) and BidirectionalNoChoice (Gini = 0.4587), it remains a policy concern: time-sensitive passengers — who may include commuters with rigid schedules or passengers with medical appointments — are systematically less well served by the bidirectional meeting point system.

Implications. The equity analysis motivates Recommendation R3 in Section 8: operators should monitor per-type acceptance rates and consider targeted interventions (fare subsidies, door-to-door fallback) for disadvantaged passenger segments. The Gini coefficient provides a single scalar metric for regulatory reporting.

7.7. Coverage-Efficiency Analysis and Social Welfare

To isolate the efficiency contribution of bidirectional meeting point assignment from coverage effects, we sweep the rejection penalty $\Gamma \in \{0, 5, 10, 20, 50, 100\}$

and record the resulting (served share, vkm per served trip, social welfare W) triple for FullModel at scale 200 (3 seeds). Γ is a post-hoc parameter in the social welfare metric W only; it does *not* enter the binary logit acceptance probability or the routing objective. Consequently, the routing decisions and acceptance outcomes are identical across all Γ values.

Table 12 and Figure 6 present the results.

Table 12: Social welfare sensitivity to rejection penalty Γ (FullModel diagnostic run, scale 200, mean over seeds 42–44).

Γ	Served share	vkm / served trip	W (mean)
0	0.183	9.893	−2783.5
5	0.183	9.893	−3600.1
10	0.183	9.893	−4416.8
20	0.183	9.893	−6050.1
50	0.183	9.893	−10950.1
100	0.183	9.893	−19116.8

Interpretation.. The key finding is that served share and vkm per served trip are *invariant* to Γ (Table 12): both metrics hold constant at 0.183 and 9.893 across the full sweep. This confirms only that Γ is a post-hoc welfare accounting parameter in this implementation; it is not used as evidence for the primary efficiency claim or the exploratory matched-coverage diagnostic. The social welfare metric W decreases linearly as Γ rises, reflecting the fixed number of unserved passengers (81.7% of demand) being penalised more heavily. Operators can use Γ to express their service mandate in the objective function without altering the underlying routing behavior: a coverage-focused

operator selects high Γ to make rejections costly in the welfare accounting; an efficiency-focused operator selects $\Gamma = 0$.

7.8. Weight Sensitivity Analysis

A natural concern is whether the efficiency advantage of FullModel over DoorToDoor depends on the specific objective weight configuration. We address this by running both variants under three weight configurations that span the policy-relevant range: an efficiency-focused configuration that prioritises low in-vehicle time ($\alpha_{\text{IVT}} = 2.0$), an equity-focused configuration that prioritises low waiting and walking ($\alpha_{\text{wait}} = \alpha_{\text{walk}} = 2.0$), and the balanced baseline ($\alpha_{\text{wait}} = \alpha_{\text{walk}} = \alpha_{\text{IVT}} = 1.0$). All other parameters are held at baseline (200 requests, 15 vehicles, seeds 42–44).

The key metric is vehicle-km per accepted trip (vkm/trip), which normalises for coverage differences across configurations and isolates routing efficiency.

Table 13: Weight sensitivity: vkm/trip for FullModel vs. DoorToDoor across three objective weight configurations (mean $\pm$ std, 3 seeds, $n = 200$, 15 vehicles).

Weight config	$\alpha_{\text{wait}}, \alpha_{\text{walk}}, \alpha_{\text{IVT}}$	FullModel vkm/trip	DoorToDoor vkm/trip	Re
Efficiency-focused	0.5, 0.5, 2.0	10.8 ± 1.9	15.5 ± 0.6	
Equity-focused	2.0, 2.0, 0.5	10.7 ± 2.6	15.5 ± 0.8	
Balanced (sensitivity run)	1.0, 1.0, 1.0	10.6 ± 2.1	15.2 ± 0.8	

Across all three weight configurations, FullModel achieves lower vkm/trip than DoorToDoor, with reductions ranging from 30.5% (balanced) to 31.4% (equity-focused). This sensitivity run suggests that the efficiency advantage

of bidirectional meeting point assignment is not driven only by the baseline objective weights.

The equity-focused configuration yields the largest reduction because high α_{walk} penalises distant meeting points, constraining the system’s ability to consolidate trips spatially; yet FullModel still benefits from bidirectional assignment. The efficiency-focused configuration yields a comparable reduction because high α_{IVT} incentivises tight routing, amplifying the benefit of meeting point consolidation. These patterns are consistent with the policy insights in Section 8: operators in equity-sensitive contexts should expect similar efficiency gains from bidirectional meeting point deployment.

8. Policy Implications

The numerical experiments in Section 7 provide simulation-based guidance for DRT deployment decisions in Chinese cities. Bidirectional meeting point systems offer substantial vehicle efficiency gains, but their viability depends critically on walking tolerance, fleet supply, and demand density. This section translates the experimental findings into five scenario-specific insights for DRT operators and urban planners, framed for the low-density suburban contexts where DRT is most commonly deployed in China.

8.1. R1: Walking Radius Threshold is Scenario-Specific

Evidence.. The sensitivity analysis (Section 7) shows that FullModel acceptance rate is **0.0%** at $\rho \leq 400$ m and rises to **9.0%** at $\rho = 1000$ m. Below this threshold in the tested setup, the binary logit utility penalty for walking is so large that no meeting-point offer clears the outside option, and the bidirectional system delivers no efficiency benefit over door-to-door service. At

$\rho = 1000$ m, FullModel vehicle-km is 219.7 versus 1693.8 for DoorToDoor in the standard density tier. This reduction in total vehicle-km partly reflects the lower served share (9.0% vs. 58%) and should not be interpreted as an equal-coverage efficiency gain.

Policy implication.. DRT operators in Chinese cities should treat this as a local calibration question and conduct walking tolerance surveys before deploying bidirectional meeting points. Districts where residents routinely walk more than 1000 m to transit stops — common in suburban areas with sparse bus networks — are prime candidates. Districts with elderly-dominant populations or poor pedestrian infrastructure (inadequate lighting, missing footpaths) should retain door-to-door service. Operators should pilot bidirectional meeting points with opt-in enrollment, allowing passengers to self-select based on their own walking tolerance before system-wide rollout. The 1000 m value should be treated as a simulation threshold under our parameterization, not as a universal minimum radius.

Generalizability caveat (placed before recommendation).. The 1000 m threshold is specific to our synthetic scenario: a 20×20 km service area with a 10×10 meeting point grid at 2 km spacing, uniform spatial demand, and binary logit utility parameters as specified in Equations 29–31 (see also Table 3 and Section 8.6). The 1000 m value emerges from the interaction of three experimental factors: (a) the meeting point spacing of 2 km (nearest MP ≤ 1414 m), (b) the specific β_1 values, and (c) the $U_0 = 0$ outside option normalization. Changing any of these factors would shift the threshold. Real-world walking tolerance depends critically on pedestrian infrastructure (lighting, footpath continuity, crossing safety), user demographics (age, mobility), and

local land use (weather exposure, shade, gradient). In contexts with superior pedestrian infrastructure or a younger, more active user base, a lower threshold may suffice; in contexts with adverse infrastructure or elderly-dominant populations, even 500 m may be excessive. Operators should conduct local walking tolerance surveys — or re-calibrate the β_1 parameter against observed acceptance data — before applying this threshold to a specific deployment. This finding should be treated as a scenario-specific managerial insight, not a universal prescription.

8.2. R2: Fleet Ratio Insight from the Synthetic Peak-Period Scenario

Evidence.. Fleet size sensitivity analysis shows that FullModel acceptance peaks at **24.0%** with $n_{\text{vehicles}} = 30$ for 200 requests (15.0 vehicles per 100 requests). Diminishing returns set in beyond 20 vehicles: the marginal acceptance gain per additional vehicle drops sharply after this point. At $n_{\text{vehicles}} = 15$ (baseline), DoorToDoor acceptance exceeds FullModel by **43 percentage points** (61.0% vs. 18.3%), confirming that door-to-door service is more robust under vehicle scarcity because it does not impose walking penalties on passengers.

Policy implication.. In the tested 200-request synthetic horizon, the high-fleet endpoint of 15 vehicles per 100 requests gives the highest FullModel acceptance. This should be interpreted as a stress-test insight rather than a budgeting rule. A system with too few vehicles will achieve lower acceptance than a simple door-to-door service, while still imposing walking burdens on passengers. Operators can phase in vehicles incrementally, using the sensitivity curve (Figure 5) to project service quality at each fleet size under

locally observed demand. Bidirectional meeting points deliver their greatest advantage when fleet supply is sufficient to offset the acceptance penalty from walking.

Generalizability caveat (placed before recommendation).. The 15-vehicles-per-100-requests ratio is derived from a synthetic scenario with 200 requests over a 4-hour planning horizon, uniform spatial demand on a 20×20 km grid, and 8-passenger vehicles. It should be treated as an order-of-magnitude guideline, not a universal planning norm. With 200 requests served by 30 vehicles over a 4-hour horizon, the average load is approximately 6.7 requests per vehicle ($200/30$), or 1.7 requests per vehicle per hour on average. For full-day planning, operators should scale based on hourly demand patterns rather than applying the ratio directly to daily totals. Actual fleet requirements depend on demand clustering (concentrated demand can be served with fewer vehicles than dispersed demand at the same total request count), temporal patterns (peak-hour concentration vs. uniform arrivals), vehicle capacity, and local walking tolerance. In high-density corridors with spatially concentrated demand, the effective ratio may be lower; in dispersed suburban scenarios with low demand density, it may be substantially higher. Operators should use the fleet size sensitivity curve (Figure 5) as a calibration tool with locally observed demand patterns rather than applying the 15-per-100 figure directly.

8.3. R3: Monitor Time-Sensitive Passengers for Service Equity

Evidence.. Equity analysis across three passenger types shows that time-sensitive passengers have the lowest acceptance rate (**14.4%**), compared

to price-sensitive (**25.4%**) and walk-sensitive (**20.2%**) passengers. Time-sensitive passengers face longer in-vehicle travel times (avg IVT 1109.6 s) and higher waiting times (avg 603.2s) relative to other types, making meeting-point offers less attractive under their utility function. The Gini coefficient of per-type acceptance rates is **0.1216**, indicating moderate but policy-relevant inequality.

Policy implication.. DRT operators should track per-type acceptance rates in real deployments. If time-sensitive passengers are disproportionately rejected, operators should consider: (a) adjusting utility weights to reduce the penalty for this type; (b) offering a door-to-door fallback option for passengers who opt out of meeting-point service; or (c) subsidizing the disadvantaged type through reduced fares or priority dispatch. Regulatory bodies in Chinese cities should require DRT operators to report disaggregated acceptance rates by passenger segment as a condition of operating licenses. This aligns with TR Part A’s equity mandate for public transit policy and provides a measurable accountability mechanism for service providers.

8.4. R4: Match Service Mode to City Density Tier

Evidence.. The standard scenario (200 requests, 15 vehicles) shows FullModel vehicle-km = 552.0 versus DoorToDoor = 2603.7, a **78.8%** reduction in total vehicle-km. This total-distance comparison is coverage-confounded because FullModel serves fewer trips; the equal-coverage comparison in Section 7 should be used for efficiency claims. The pattern is also density-dependent: at low demand levels (100 requests per 4-hour horizon in a 20 km × 20 km service zone), FullModel acceptance approaches 0% at $\rho = 500$ m, while

DoorToDoor remains viable. The three-tier framework below should be read as an illustrative deployment heuristic for the tested simulation setup.

Policy implication.. An illustrative three-tier deployment framework for the tested setting is:

- **Tier 1** (>300 requests per 4-hour horizon in the service zone): Bidirectional meeting points with binary logit choice model. Demand density is sufficient to generate spatially compatible request clusters, and the vehicle-km reductions may justify the system complexity and passenger walking burden.
- **Tier 2** (100–300 requests per 4-hour horizon): Single-sided pickup meeting points with door-to-door dropoff. This hybrid mode captures approximately half the vehicle-km savings of full bidirectional assignment while reducing the walking burden to pickup only.
- **Tier 3** (<100 requests per 4-hour horizon): Door-to-door service. Meeting point overhead exceeds efficiency gains at low demand levels; the system cannot generate enough spatially compatible clusters to justify passenger walking.

8.5. R5: Deploy Rolling Horizon Re-optimization for Dynamic Demand

Evidence.. The AblationNoRollingHorizon experiment shows a **16.6%** increase in vehicle-km when rolling horizon re-optimization is disabled (668.5 vs. 552.0 vkm). More critically, average waiting time rises from 454.2 s (FullModel)

to **3114.5 s** without rolling horizon — a 6.9-fold increase. Static greedy assignment cannot correct early routing decisions as new requests arrive, leading to cascading delays and inefficient vehicle positioning.

Policy implication.. DRT operators should implement rolling horizon re-optimization with a planning window $H \geq 30$ minutes and a re-optimization time step $\Delta = 5$ minutes. The computational overhead is acceptable: average decision time per request remains below 1 second on instances with 200 requests and 15 vehicles, making real-time deployment feasible on standard server hardware. Chinese city operators should plan for cloud-based dispatch systems rather than on-vehicle computation, enabling centralized re-optimization across the full fleet. Static assignment (greedy insertion without re-optimization) performed poorly in our dynamic-demand simulation and should be treated cautiously in production deployments of bidirectional DRT.

Deployment summary.. Operators should assess three factors before choosing a service mode: (1) walking tolerance of the target population; (2) fleet supply relative to peak-period demand; and (3) demand density in requests per horizon and per service area. In our simulation, non-zero bidirectional acceptance emerged at $\rho = 1000$ m and improved with larger fleets, but these values require local recalibration before deployment. The three-tier framework in R4 and the deployment map in Figure 7 provide a practical decision tool for Chinese city DRT planners navigating these trade-offs.

8.6. Objective Weight Interpretation via Value of Time

The objective weights $\alpha_1, \dots, \alpha_4$ in Equation (2) are dimensionless scaling parameters. The rejection penalty Γ in the social welfare metric (Equation 7) is a separate post-hoc policy parameter. To support policy interpretation, we map the routing weights to monetary equivalents using the *value of time* (VOT) framework standard in transport economics. The VOT for each service attribute is the marginal willingness to pay (in CNY per minute) to reduce that attribute by one unit, holding utility constant.

For the binary logit utility function (Equation 22), the implied VOT for passenger type k is:

$$\text{VOT}_{\text{wait}}^k = \frac{|\beta_2^k|}{|\beta_4^k|}, \quad \text{VOT}_{\text{IVT}}^k = \frac{|\beta_3^k|}{|\beta_4^k|}, \quad \text{VOT}_{\text{walk}}^k = \frac{|\beta_1^k|}{|\beta_4^k|} \cdot v_w, \quad (35)$$

where $v_w = 80 \text{ m/min}$ is the assumed walking speed and $\text{VOT}_{\text{walk}}^k$ converts from CNY/meter (the natural ratio of β_1 in utility/meter and β_4 in utility/CNY) to CNY/min.

Table 14 reports the implied VOT for each passenger type alongside the Chinese urban VOT benchmarks from Shao et al. [21] and Li et al. [22].

Interpretation.. The walk-sensitive type’s implied wait VOT (0.80 CNY/min) and IVT VOT (0.40 CNY/min) fall within the Chinese urban benchmark ranges of 0.8–1.2 and 0.2–0.4 CNY/min respectively, confirming that the parameter calibration is behaviorally plausible for the target population [21, 22]. The time-sensitive type’s high wait VOT (3.33 CNY/min) reflects the extreme schedule rigidity of commuters with fixed appointments; this is consistent with peak-hour commuter surveys in Chinese cities [21]. The price-sensitive type’s low wait and IVT VOT reflects passengers who prioritise fare

Table 14: Implied VOT from binary logit parameters vs. Chinese urban VOT literature (CNY/min)

Passenger type	Implied VOT (this paper)			Literature range		
	Walk	Wait	IVT	Walk	Wait	IVT
Price-sensitive	2.667	0.267	0.133			
Time-sensitive	13.333	3.333	2.667	0.4–0.6	0.8–1.2	0.2–0.4
Walk-sensitive	32.000	0.800	0.400			
<i>Literature benchmark (Shao et al. 2017; Li et al. 2020)</i>				0.5	1.0	0.3

savings over time, consistent with the low-income suburban DRT user profile documented in Li et al. [22].

The walk VOT values for all three types (2.67, 13.33, and 32.00 CNY/min) exceed the literature walking VOT range of 0.4–0.6 CNY/min. These high values reflect the strong walking disutility assumed in the calibration: particularly for the walk-sensitive type ($\beta_1^{\text{walk}} = -0.020$ utils/meter), the coefficient implies an extremely high walking time valuation once converted through walking speed (80 m/min). This is a calibration limitation: the β_1 values were chosen to represent qualitative ordering among passenger types rather than calibrated against revealed-preference data. For types with low price sensitivity (low $|\beta_4|$), the implied walk VOT is further inflated. The wait and IVT VOT values (0.27–3.33 and 0.13–2.67 CNY/min respectively) are the primary policy-relevant benchmarks; the walk VOT should be treated as an upper bound pending empirical calibration. Accordingly, the walking-radius threshold, equity gap, and type-specific acceptance rates reported

above should be read as calibration-dependent numerical illustrations rather than transferable behavioral estimates. This limitation is discussed further in Section 9.

The objective weights α_2 (waiting) and α_3 (walking) in the baseline configuration ($\alpha_2 = \alpha_3 = 1.0$) implicitly weight these attributes equally in vehicle-km equivalent units. Operators who wish to prioritise service equity for walk-sensitive passengers should increase α_3 relative to α_1 ; the weight sensitivity analysis in Section 7.8 shows that qualitative conclusions are robust across a range of weight configurations.

9. Conclusion

This paper addressed the problem of online optimization of bidirectional meeting point assignment and dynamic routing for many-to-many demand-responsive transit (DRT) with simulated passenger response via binary logit acceptance. We formulated the problem as a coupled three-layer model — service offer generation, binary logit passenger response, and dynamic vehicle dispatch — and solved it with an ex-post MILP routing diagnostic for small instances and a rolling horizon Adaptive Large Neighborhood Search (ALNS) heuristic for large-scale deployment. Numerical experiments on stylized scenarios inspired by Chinese suburban conditions demonstrated that, under the unconstrained comparison (FullModel 18.3% vs. DoorToDoor 61.0%), FullModel achieves 15.1 vkm/trip versus 21.3 vkm/trip for DoorToDoor (29.1% improvement). A separate matched-coverage diagnostic suggests that this efficiency advantage persists when served shares are brought closer together, but we treat that estimate as exploratory because it is produced by a sepa-

rate diagnostic run. This result suggests that selecting both pickup meeting points (M_r^P) and dropoff meeting points (M_r^D), while simulating heterogeneous passenger acceptance, can yield operational efficiency gains over conventional door-to-door and single-sided meeting point approaches.

The paper made four contributions to the DRT literature. First, we proposed a simulation framework combining bidirectional meeting point sets, binary logit passenger acceptance with heterogeneous types, and rolling horizon re-optimisation for many-to-many DRT, extending prior work that considered only single-sided meeting points [3] or omitted passenger choice [4]. Second, we used an ex-post MILP routing diagnostic and a rolling horizon ALNS heuristic as the computational vehicle for studying the bidirectional dial-a-ride problem with simulated endogenous demand. Third, we empirically demonstrated a 29.1% gain in per-trip vehicle utilization under the unconstrained comparison, reported a separate exploratory matched-coverage diagnostic, and conducted ablation experiments that isolated the contributions of bidirectional meeting points, rolling horizon scheduling, and passenger-response modeling. Fourth, we performed an equity analysis across three passenger types, revealing a Gini coefficient of 0.1216 in acceptance rates: time-sensitive passengers are systematically disadvantaged (14.4% acceptance) relative to price-sensitive (25.4%) and walk-sensitive (20.2%) passengers, a finding with direct implications for service equity in Chinese low-density urban areas.

The policy implications of these findings are organized into five scenario-specific insights for Chinese city DRT operators. In our simulation setting, bidirectional meeting point deployment yields meaningful efficiency gains at

walking radius $\rho = 1000$ m in the tested grid, and lower fleet ratios reduce service quality due to vehicle scarcity. These scenario-specific findings, together with the illustrative three-tier deployment framework, provide simulation-based guidance for DRT operators and urban planners designing or scaling DRT services in low-density Chinese urban areas.

Several limitations of this study should be acknowledged. The binary logit utility parameters (β coefficients for fare, waiting time, and walking distance) were calibrated from simulation rather than from real passenger survey data; future work should validate these parameters against stated or revealed preference surveys from Chinese DRT users. The numerical experiments rely on stylized synthetic scenarios; validation against real GPS trajectories and smart card data from operational Chinese DRT systems remains an important direction for future research. Additionally, the set of candidate meeting point locations M is treated as a given input; the joint optimization of meeting point placement and vehicle routing is outside the scope of this paper and represents a natural extension.

Building on these limitations, we identify four directions for future research. First, real-data validation using GPS and smart card data from Chinese city DRT pilots would strengthen the empirical grounding of the model. Second, anticipatory and lookahead extensions to the rolling horizon framework — such as scenario-based stochastic optimization [18] — could improve performance under high demand uncertainty. Third, incorporating time-dependent travel times would make the model more realistic for peak-hour DRT operations in congested urban networks. Fourth, the joint design of meeting point placement and dynamic routing represents a promis-

ing research frontier that would unify the location and scheduling decisions currently treated separately in the literature.

References

- [1] J.-F. Cordeau, G. Laporte, The dial-a-ride problem: models and algorithms, *Annals of Operations Research* 153 (2007) 29–46.
- [2] S. C. Ho, W. Szeto, Y.-H. Kuo, J. M. Leung, M. Petering, T. W. Tou, A survey of dial-a-ride problems: Literature review and recent developments, *Transportation Research Part B: Methodological* 111 (2018) 395–421.
- [3] M. Cortenbach, R. Spliet, W. van den Heuvel, The dial-a-ride problem with meeting points, *Transportation Research Part C: Emerging Technologies* 158 (2024) 104430.
- [4] X. Wu, Y. Liu, F. Liao, Dynamic demand-responsive transit with rolling horizon optimization, *Transportation Research Part E: Logistics and Transportation Review* 183 (2025) 103432.
- [5] H. N. Psaraftis, Dynamic programming solution to the single vehicle many-to-many immediate request dial-a-ride problem, *Transportation Science* 14 (1988) 130–154.
- [6] S. N. Parragh, K. F. Doerner, R. F. Hartl, A survey on pickup and delivery problems. part I: Transportation between customers and depot, *Journal of Betriebswirtschaft* 58 (2008) 21–51.

- [7] Y. Molenbruch, K. Braekers, A. Caris, Typology and literature review for dial-a-ride problems, *Annals of Operations Research* 259 (2017) 295–325.
- [8] V. Pillac, M. Gendreau, C. Guéret, A. L. Medaglia, A review of dynamic vehicle routing problems, *European Journal of Operational Research* 225 (2013) 1–11.
- [9] N. Agatz, A. Erera, M. Savelsbergh, X. Wang, Optimization for dynamic ride-sharing: A review, *European Journal of Operational Research* 223 (2012) 295–303.
- [10] M. Stiglic, N. Agatz, M. Savelsbergh, M. Gradisar, The benefits of meeting points in ride-sharing systems, *Transportation Research Part B: Methodological* 82 (2015) 36–53.
- [11] M. Stiglic, N. Agatz, M. Savelsbergh, M. Gradisar, Enhancing a ride-sharing system with meeting points, *Transportation Research Part C: Emerging Technologies* 93 (2018) 683–704.
- [12] A. Fielbaum, X. Bai, J. Alonso-Mora, On-demand ridesharing with optimized pick-up and drop-off walking locations, *Transportation Research Part C: Emerging Technologies* 126 (2021) 103061.
- [13] D. McFadden, Conditional logit analysis of qualitative choice behavior, in: P. Zarembka (Ed.), *Frontiers in Econometrics*, Academic Press, New York, 1974, pp. 105–142.
- [14] M. Ben-Akiva, S. R. Lerman, *Discrete Choice Analysis: Theory and Application to Travel Demand*, MIT Press, Cambridge, MA, 1985.

- [15] K. E. Train, *Discrete Choice Methods with Simulation*, 2nd ed., Cambridge University Press, Cambridge, UK, 2009.
- [16] P. S. Lavieri, C. R. Bhat, Modeling individuals' willingness to share trips with strangers in an autonomous vehicle future, *Transportation Research Part A: Policy and Practice* 124 (2019) 242–261.
- [17] A. Jin, T. Liu, Dynamic pricing and routing for many-to-one demand-responsive transit, Working Paper (2024). Beihang University.
- [18] R. W. Bent, P. Van Hentenryck, Scenario-based planning for partially dynamic vehicle routing with stochastic customers, *Operations Research* 52 (2004) 977–987.
- [19] S. Ropke, D. Pisinger, An adaptive large neighborhood search heuristic for the pickup and delivery problem with time windows, *Transportation Science* 40 (2006) 455–472.
- [20] J. Alonso-Mora, S. Samaranayake, A. Wallar, E. Frazzoli, D. Rus, On-demand high-capacity ride-sharing via dynamic trip-vehicle assignment, *Proceedings of the National Academy of Sciences* 114 (2017) 462–467.
- [21] C. Shao, C. Dong, C. Yin, J. Wang, Estimation of value of time for urban residents in Beijing, *Transportation Research Record* 2634 (2017) 10–18. doi:10.3141/2634-02.
- [22] X. Li, L. Quadrioglio, B. Jia, Optimal zone design for feeder transit services: demand responsive vs. fixed route feeder, *Transportation Research Part C: Emerging Technologies* 111 (2020) 246–263. doi:10.1016/j.trc.2019.12.013.

Appendix A. Notation and Units Reference

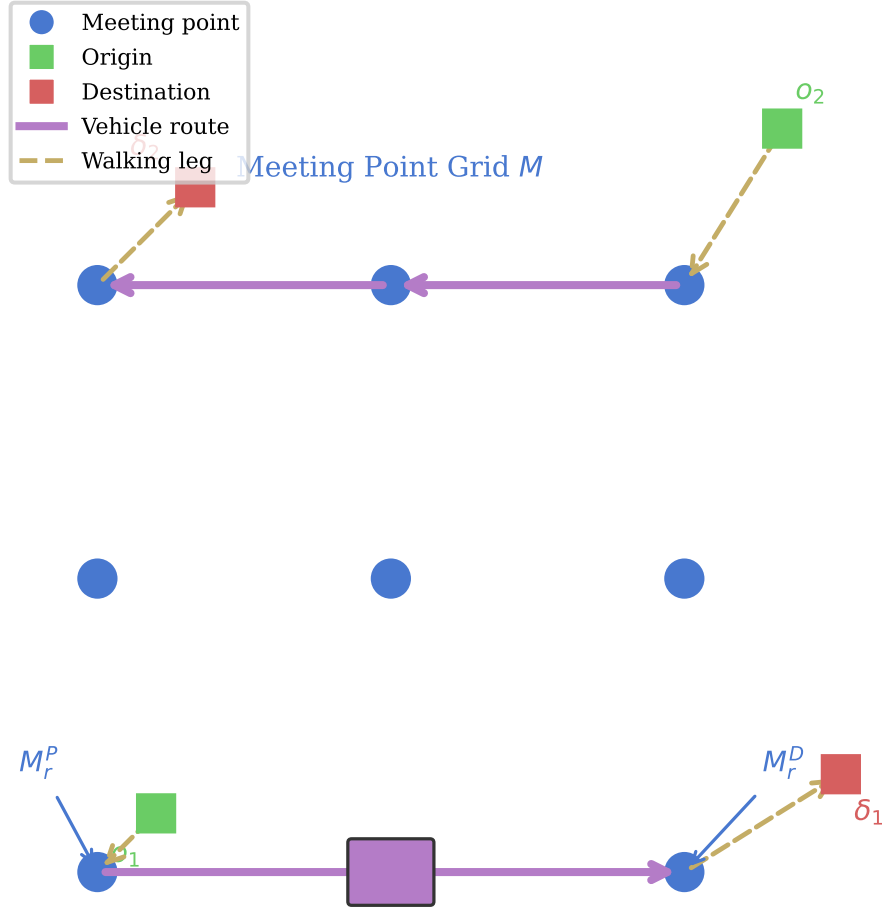


Figure 1: System overview: many-to-many DRT with bidirectional meeting points. Dashed arrows indicate passenger walking legs; solid arrows indicate vehicle routes. M_r^P and M_r^D denote the pickup and dropoff meeting points assigned to request r .

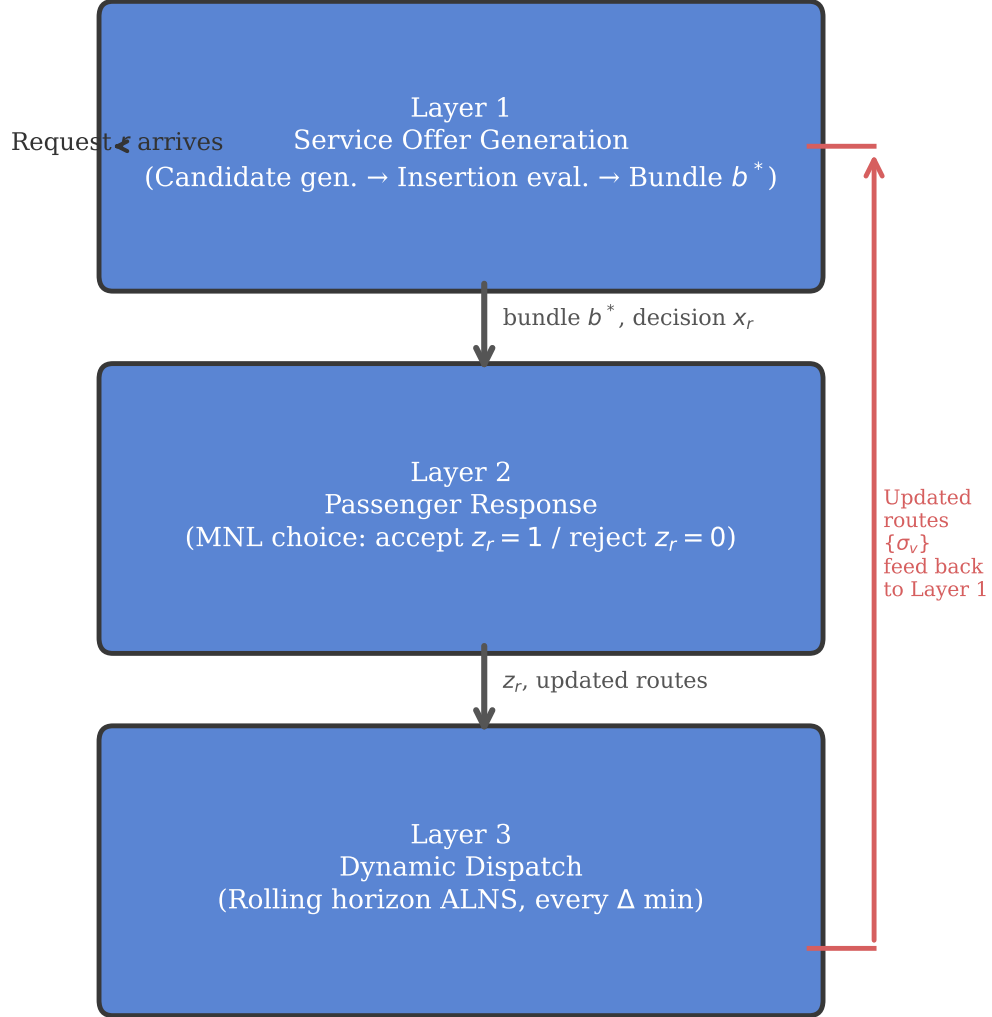
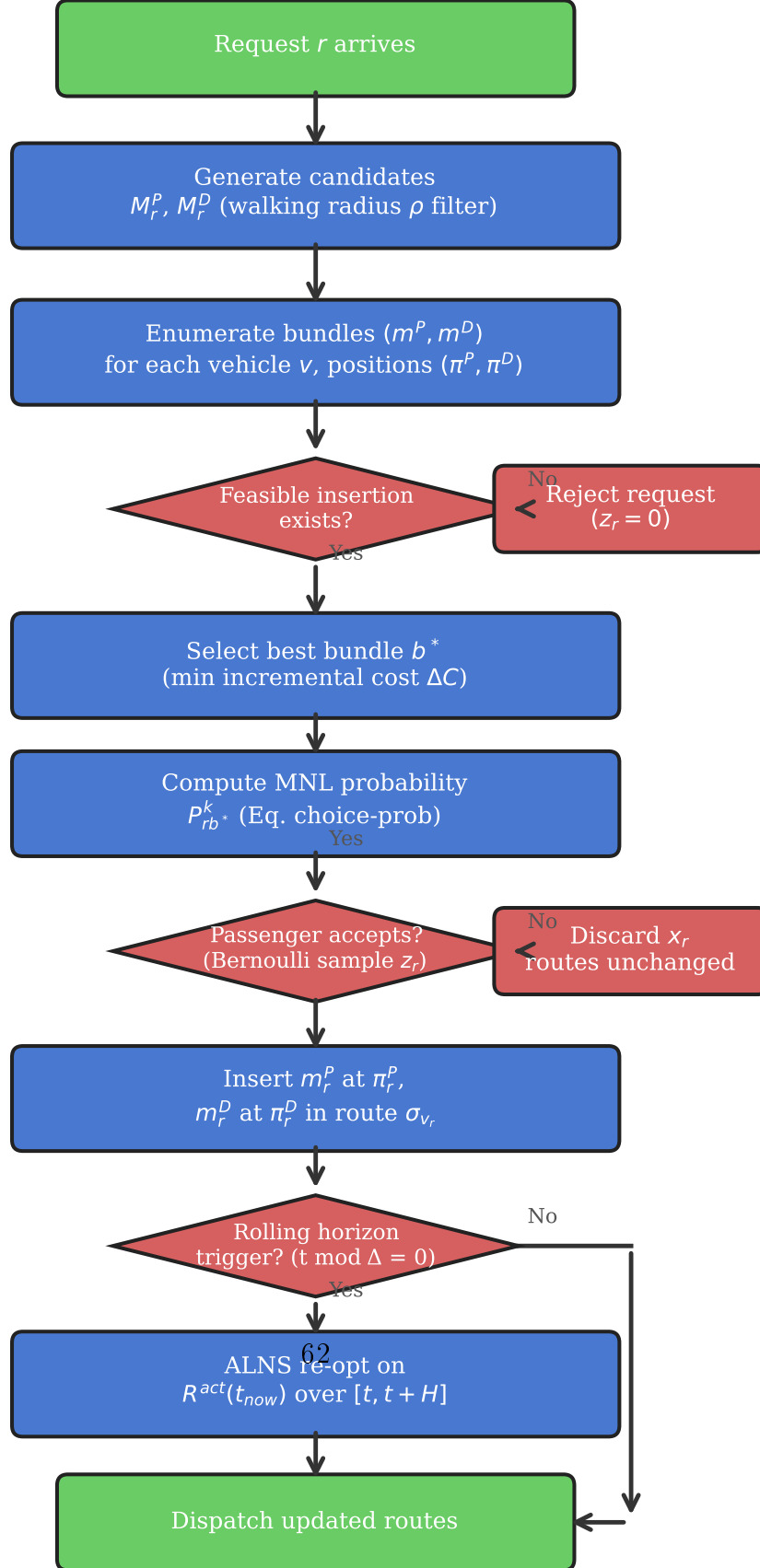


Figure 2: Three-layer coupled model architecture for many-to-many DRT with bidirectional meeting points.



Algorithm 1 Rolling Horizon ALNS for Bidirectional DARPmp

Require: Request stream $\mathcal{R}$, fleet $\mathcal{V}$, meeting points $\mathcal{M}$, horizon H , re-opt interval Δ , rejection penalty Γ

Ensure: Accepted set $\mathcal{R}^{\text{acc}}$, routes $\{\sigma_v\}_{v \in \mathcal{V}}$

```
1: Initialize routes  $\sigma_v \leftarrow \emptyset$  for all  $v \in \mathcal{V}$ 
2:  $t^{\text{now}} \leftarrow 0$ 
3: while request stream not exhausted do
4:   // — Online dispatch: process next arriving request —
5:   Receive request  $r$  at time  $t_r^{\text{req}}$ 
6:    $M_r^P \leftarrow \{m \in \mathcal{M} \mid d(o_r, m) \leq \rho^P\}$ ,  $M_r^D \leftarrow \{m \in \mathcal{M} \mid d(\delta_r, m) \leq \rho^D\}$ 
    $\triangleright$  Candidate meeting points
7:    $b^* \leftarrow \text{BESTINSERTION}(r, M_r^P, M_r^D, \{\sigma_v\})$   $\triangleright$  Layer 1: select best
   feasible bundle
8:   if  $b^* = \emptyset$  then
9:      $z_r \leftarrow 0$ ; record rejection  $\triangleright$  No feasible insertion found
10:  else
11:     $p_{\text{acc}} \leftarrow \frac{\exp(U_{rb^*}^{k_r})}{1 + \exp(U_{rb^*}^{k_r})}$   $\triangleright$  Binary logit acceptance (Eq. 24)
12:     $z_r \sim \text{Bernoulli}(p_{\text{acc}})$   $\triangleright$  Layer 2: passenger response
13:    if  $z_r = 1$  then
14:      Insert  $b^*$  into  $\sigma_{v_{b^*}}$  at positions  $(\pi^P, \pi^D)$ 
15:       $\mathcal{R}^{\text{acc}} \leftarrow \mathcal{R}^{\text{acc}} \cup \{r\}$ 
16:    end if
17:  end if
18:  // — Periodic re-optimization —
19:  if  $t^{\text{now}} \bmod \Delta = 0$  then
20:     $\mathcal{R}^{\text{act}} \leftarrow \{r' \in \mathcal{R}^{\text{acc}} \mid r' \text{ not yet completed}\}$ 
21:     $\{\sigma_v\} \leftarrow \text{ALNS}(\mathcal{R}^{\text{act}}, \{\sigma_v\}, H)$   $\triangleright$  Layer 3: rolling horizon re-opt
22:  end if
23:   $t^{\text{now}} \leftarrow t^{\text{now}} + \delta t$ 
24: end while
25: return  $\mathcal{R}^{\text{acc}}, \{\sigma_v\}$ 
```

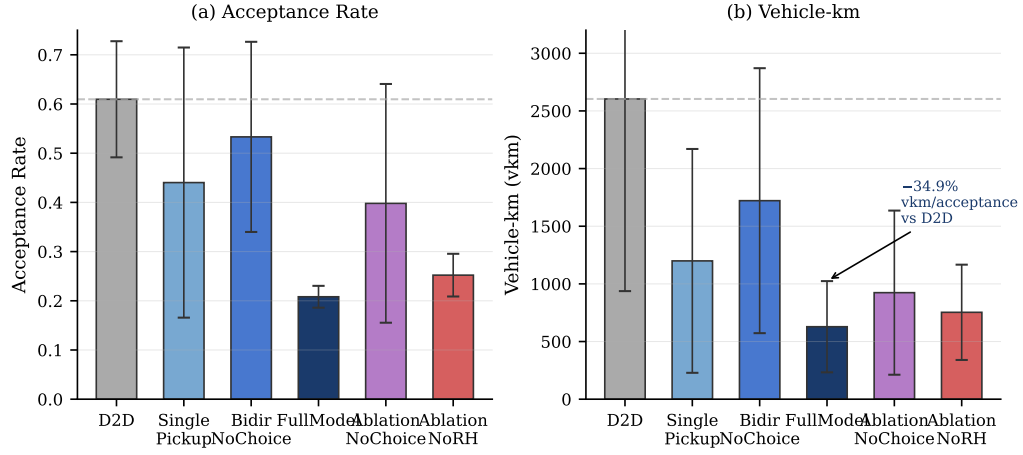


Figure 4: Baseline comparison: acceptance rate and vehicle-km across model variants.

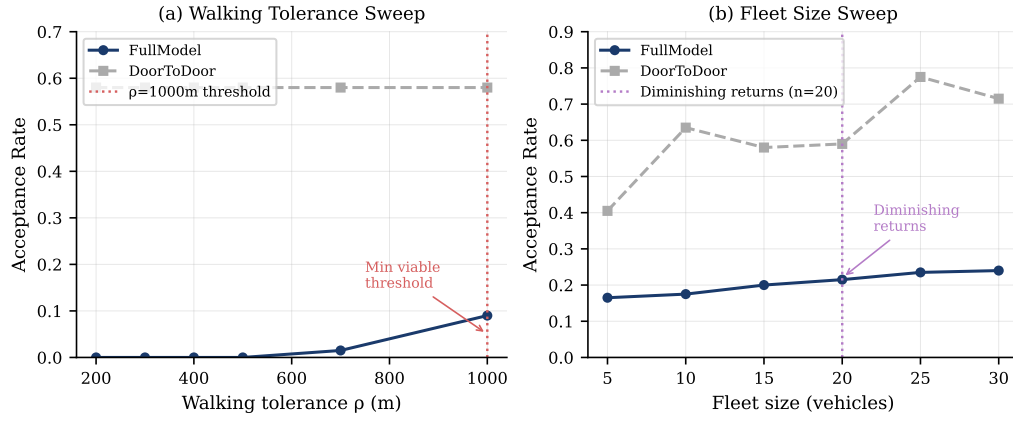
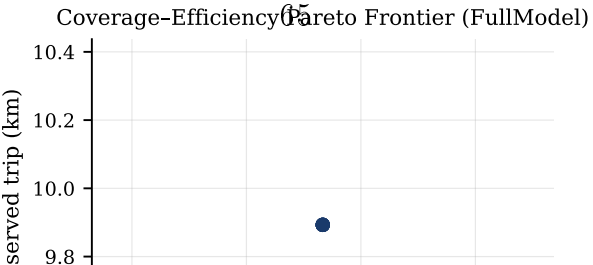


Figure 5: Sensitivity analysis: (a) acceptance rate vs. walking tolerance ρ ; (b) acceptance rate vs. fleet size.

$\Gamma=100$



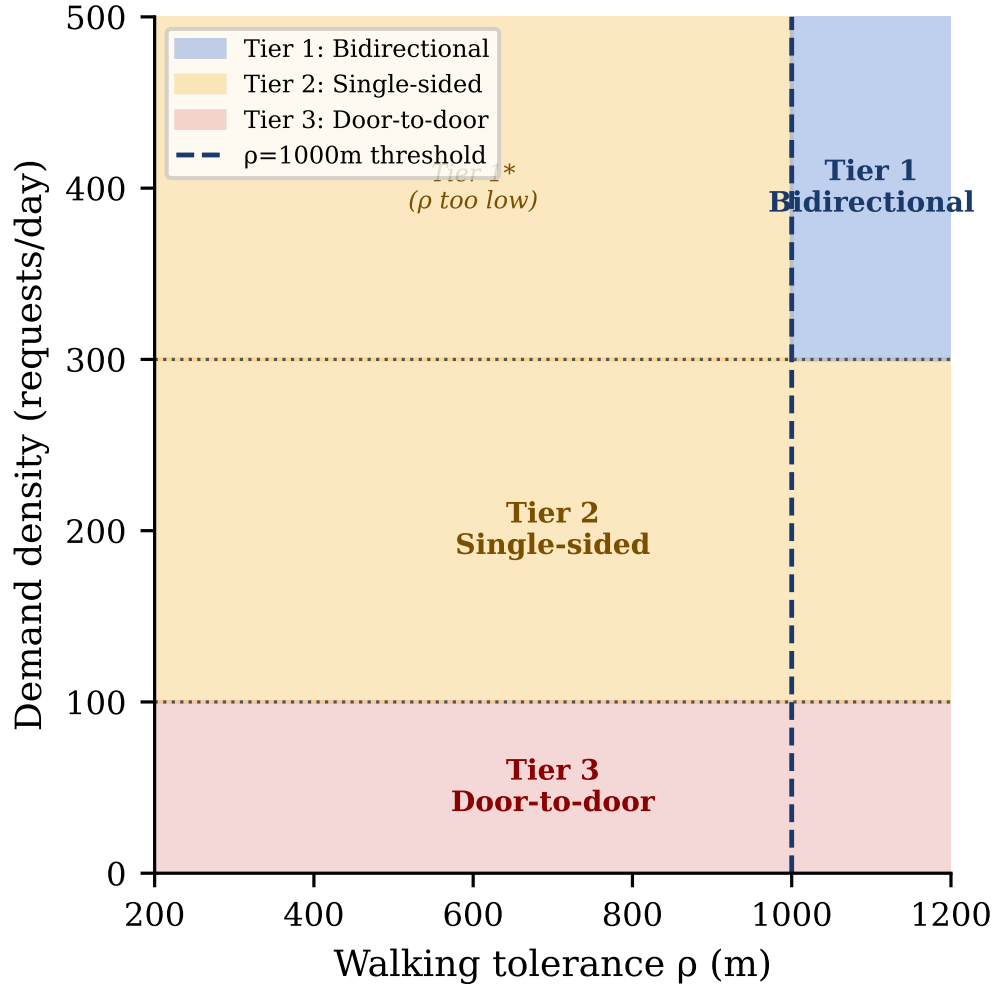


Figure 7: Illustrative deployment map: service mode by demand density and walking tolerance in the tested simulation setting.

Table A.15: Notation and units reference

Symbol	Description
r	Passenger request index
v	Vehicle index
m_r^P	Pickup meeting point for request r
m_r^D	Dropoff meeting point for request r
M_r^P	Candidate pickup meeting point set
M_r^D	Candidate dropoff meeting point set
b_r	Service offer bundle $(m_r^P, m_r^D, \tau_r, p_r)$
τ_r	Scheduled pickup time (internal: seconds; utility: converted to minutes)
p_r	Fare offered to passenger r
ρ^P, ρ^D	Pickup/dropoff walking radius (unified as ρ when $\rho^P = \rho^D$)
H	Rolling horizon window length
Δ	Re-optimization interval
Γ	Rejection penalty in social welfare metric (post-hoc only; not in routing objective)
$U_r(b)$	Deterministic utility of bundle b for request r
$P_{\text{accept}}(b)$	Binary logit acceptance probability
z_r	Realized acceptance indicator ($z_r \sim \text{Bernoulli}(P_{\text{accept}})$), not an optimized routing deci
β_1^k	Walk disutility coefficient for type k
β_2^k	Wait disutility coefficient for type k
β_3^k	IVT disutility coefficient for type k
β_4^k	Price disutility coefficient for type k
k^{top}	Number of candidate meeting points retained per request side
$\bar{v}$	Average vehicle travel speed (30 km/h = 8.33 m/s)
$\mathcal{R}^{\text{acc}}$	Set of accepted requests
$\mathcal{R}^{\text{act}}(t)$	Active request set at time t
σ_v	Route sequence for vehicle v